





Anatomy & Histology of Endocrine Glands









Chapter 13 - Endocrine Glands

The **endocrine system** is composed of glands that synthesize and secrete products, called **hormones**, directly into the blood rather than through a duct. Hormones are transported throughout the body where they influence only those cells that have receptors for that hormone.

Hormones can be:

- Proteins/peptides (e.g., insulin, growth hormone)
- Lipid derivatives - derived from cholestrol or fatty acids (*e.g.*, steroids, eicosanoids)
- Amino acid derivatives - derived from tyrosine or tryptophan (*e.g.*, epinephrine, melatonin, thyroid hormones)

Hormone effects are described as:

- **Endocrine** - released into blood and act at long distances
- **Paracrine** - diffuse a short distance through tissue fluids to nearby cells
- **Autocrine** - act on the same cell that produced the hormone

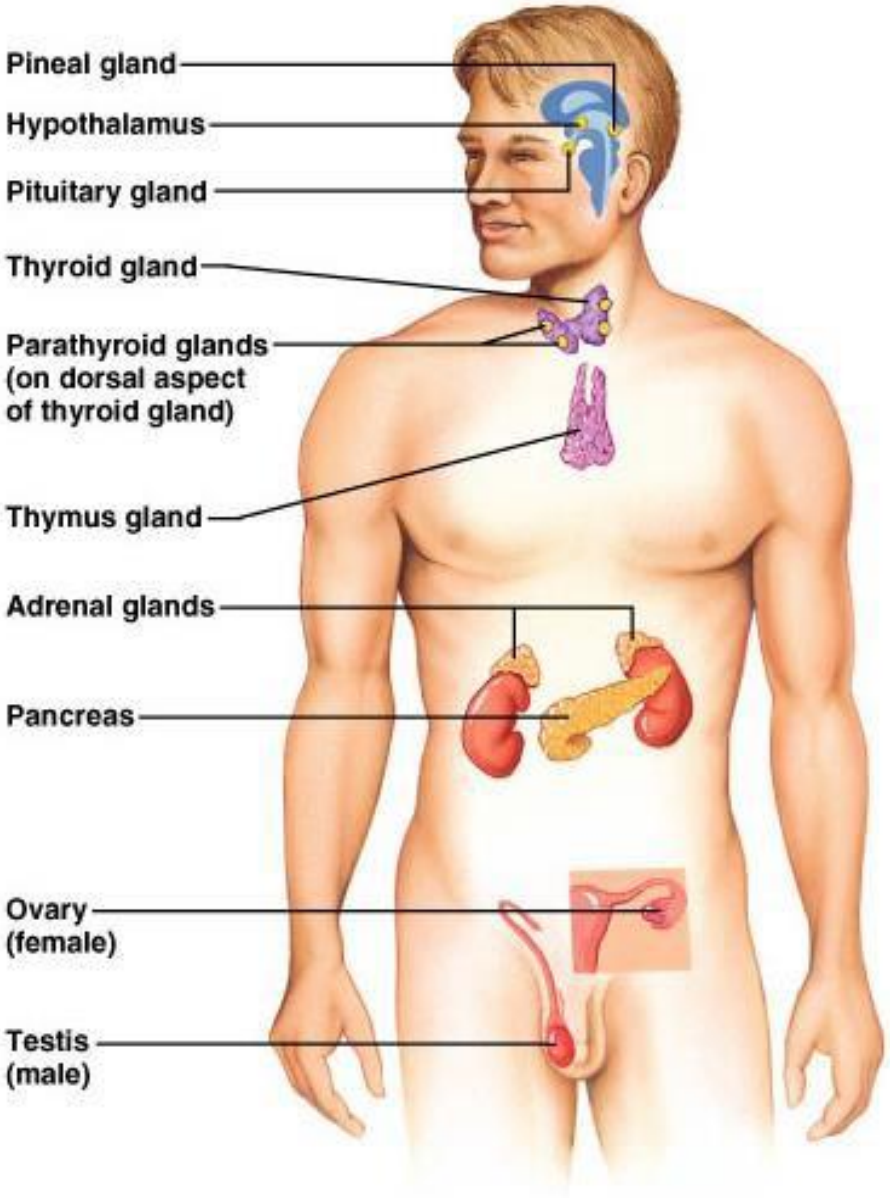


Fig 006 Endocrine Glands

Endocrine glands are highly vascular and often contain fenestrated capillaries to facilitate the diffusion of hormones into blood.

CELL AND TISSUES
1 The Cell
2 Epithelium
3 Connective Tissue
4 Muscle Tissue
5 Cartilage and Bone
6 Nervous Tissue
7 Peripheral Blood
8 Hematopoiesis
ORGAN SYSTEMS
9 Cardiovascular System
10 Lymphoid System
11 Skin
12 Exocrine Glands
13 Endocrine Glands
14 Gastrointestinal Tract
15 Liver and Gallbladder
16 Urinary System
17 Respiratory System
18 Female Reproductive

http://www.histologyguide.com/slidebox/13-endocrine-glands.html?fbclid=IwAR1u6r11MgfgdQp8Mp7b589ctuLQo8ABwY9KGLkRDIBerj9_i9pz9oIa-tk

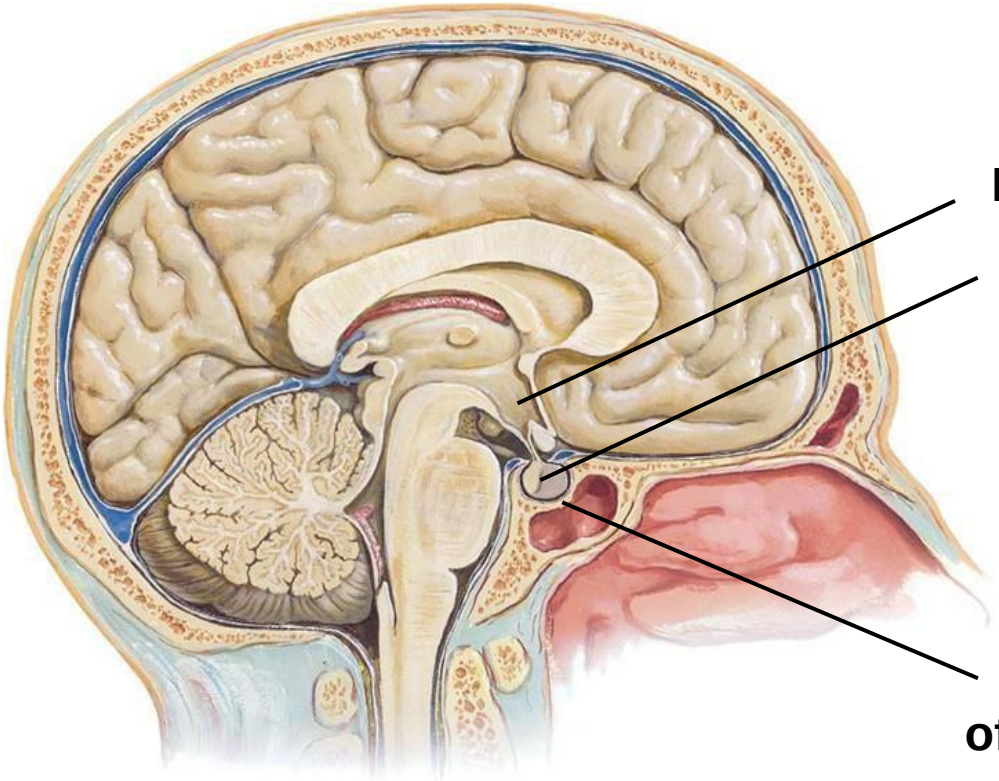


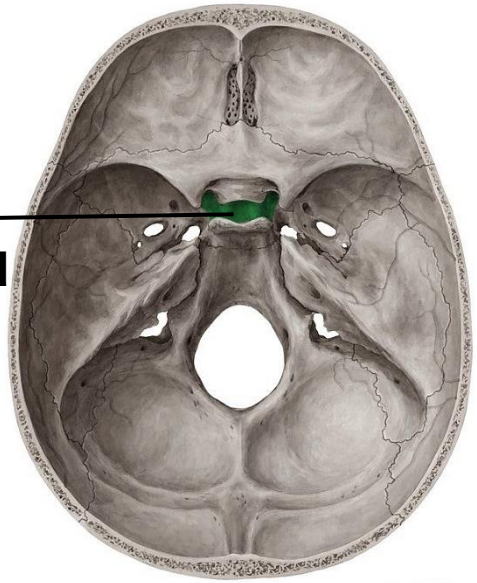
Learning objectives:

1. Name location, & blood supplies : Pituitary, thyroid, parathyroid and adrenal glands
2. List the secretory cells
3. Describe the basic functions of each hormone

I. Pituitary gland

The **pituitary gland** lies at the middle of the base of the skull and is located in a bony structure called the sella turcica, which is behind the nose and immediately beneath the hypothalamus.



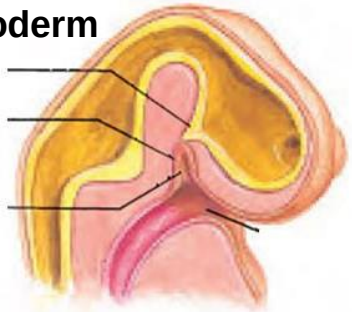


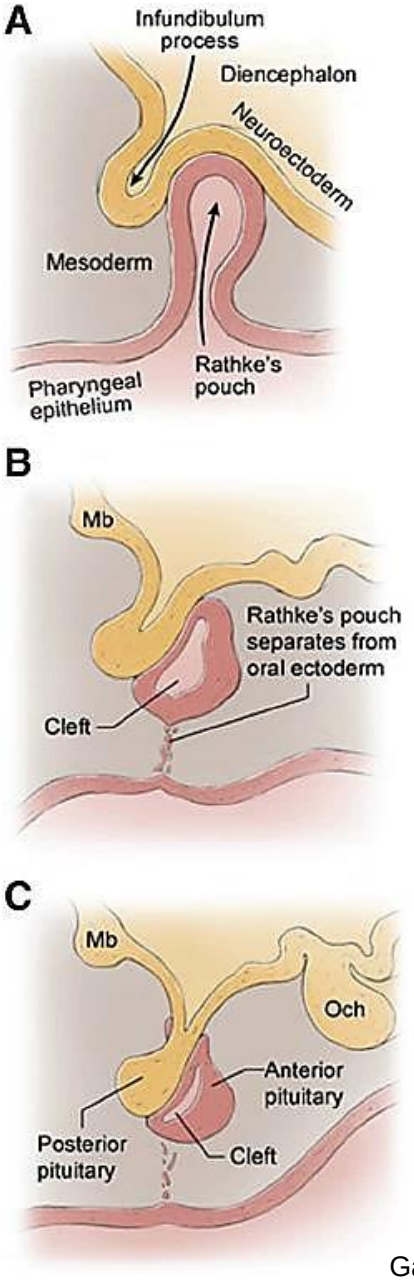
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Development of the pituitary gland

The pituitary gland has **two** anatomically & functionally separate portions: **Anterior lobe & Posterior lobe**

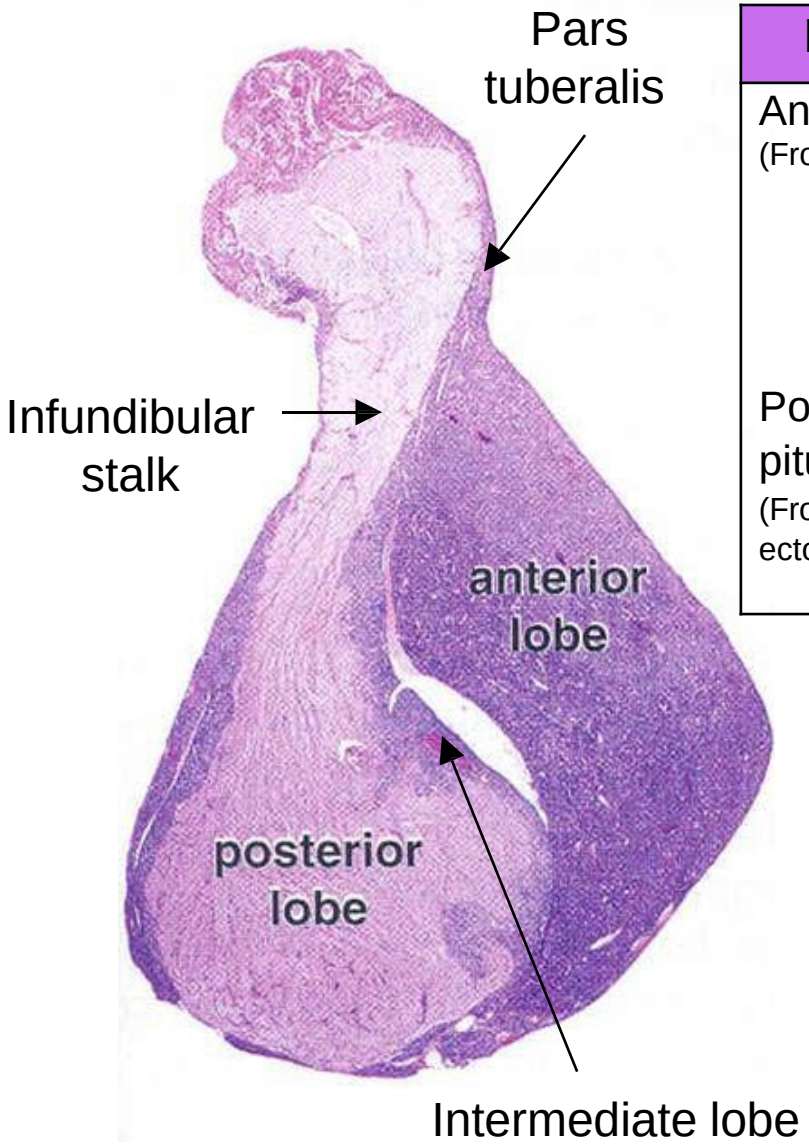






Gartner, L.P. and Hiatt, J.L. 2005. Color Atlas of Histology. 5th ed)

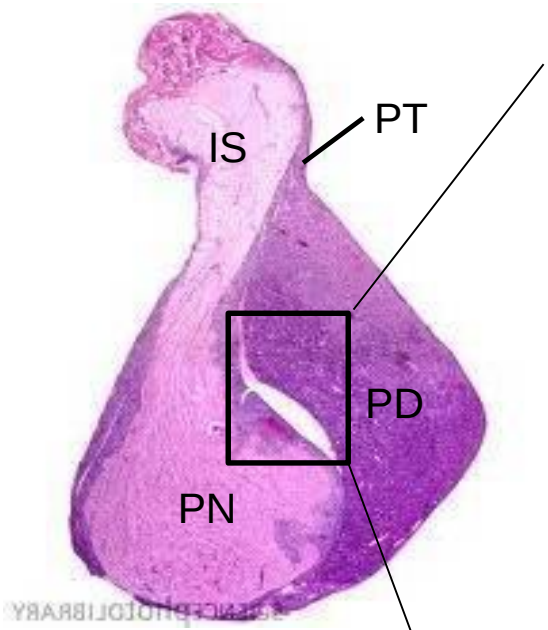
Divisions of the pituitary



Main division	Subdivision
Anterior pituitary (From oral ectoderm)	Pars distalis (anterior lobe) Pars tuberalis Pars intermedia (Intermediate lobe)
Posterior pituitary (From neural ectoderm)	Pars nervosa (posterior lobe) Infundibular stalk Median eminence

http://www.histology-world.comphotoalbumdisplayimage.

Histology of the pituitary

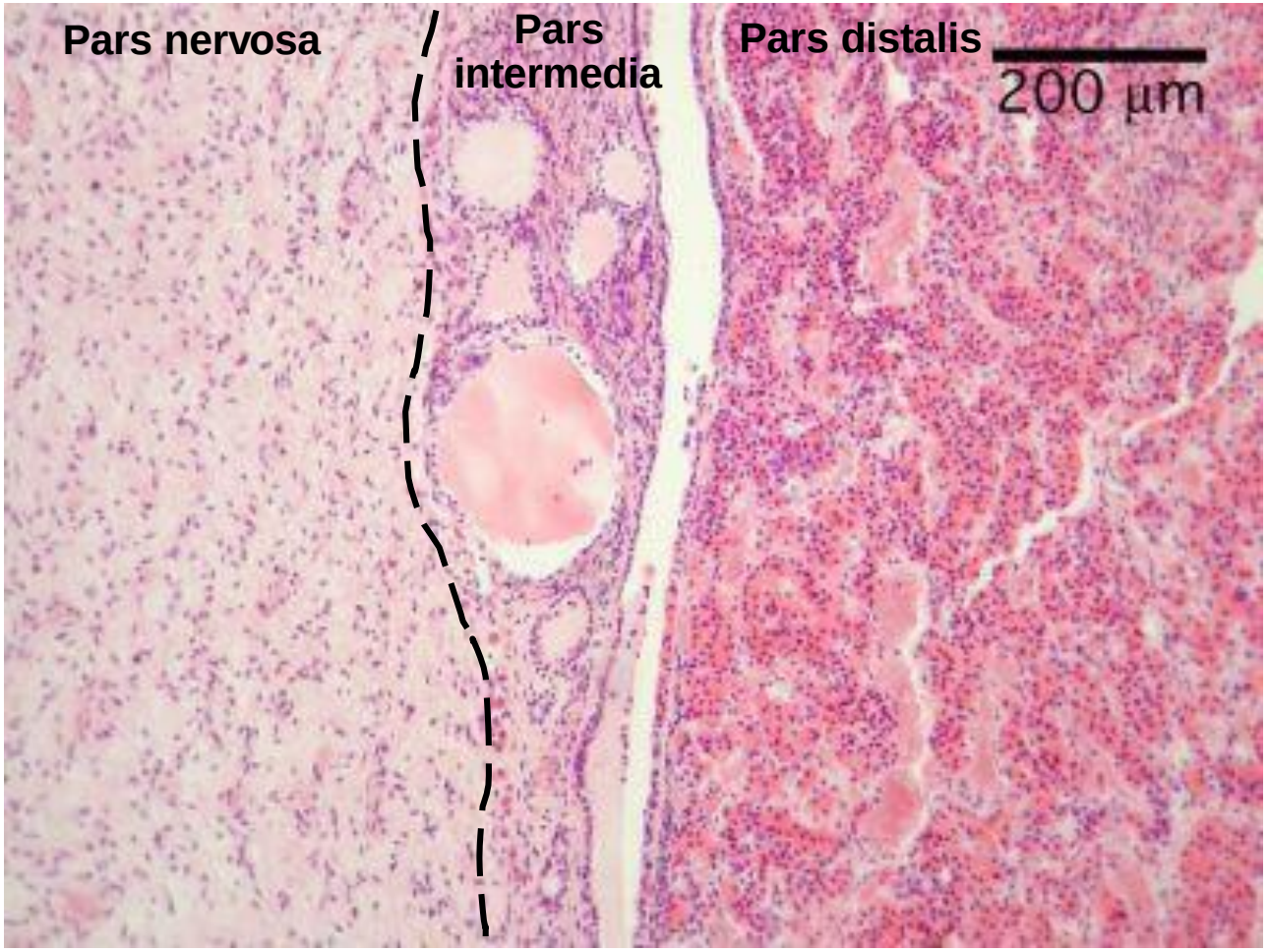


IS=Infundibular stalk

PT= Pars tuberalis

PD=Pars distalis

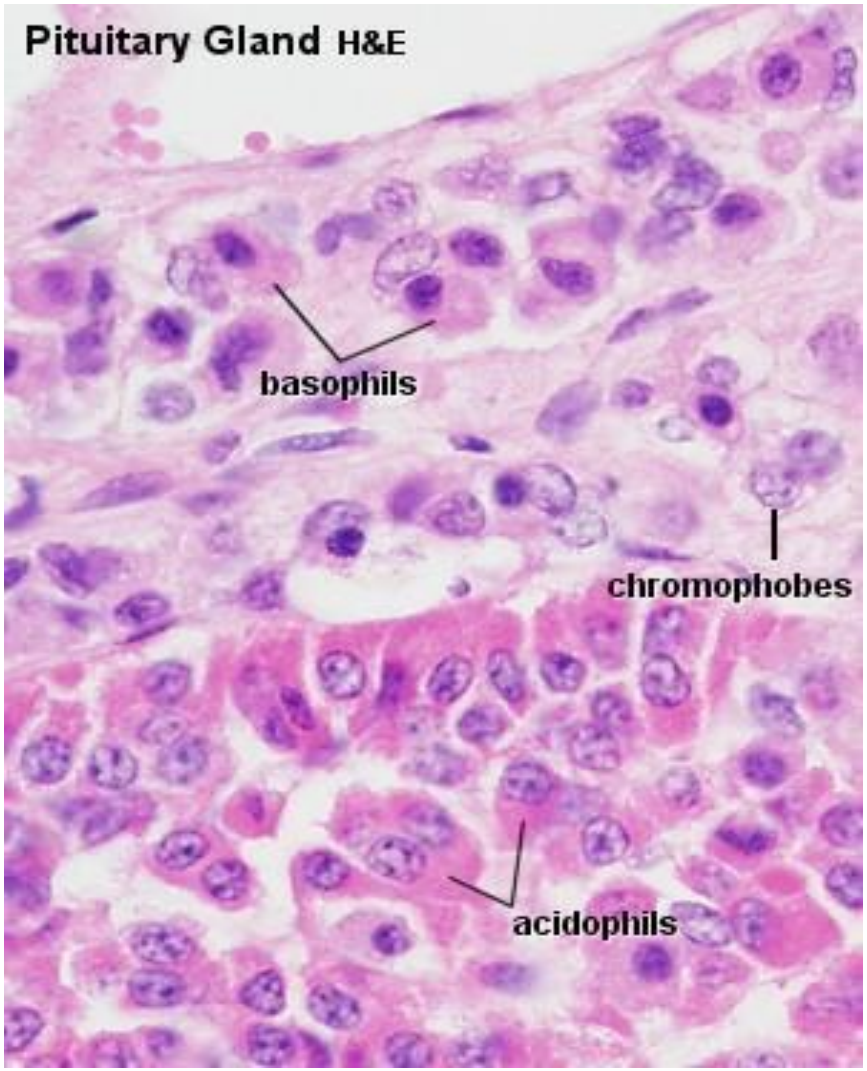
PN=Pars nervosa

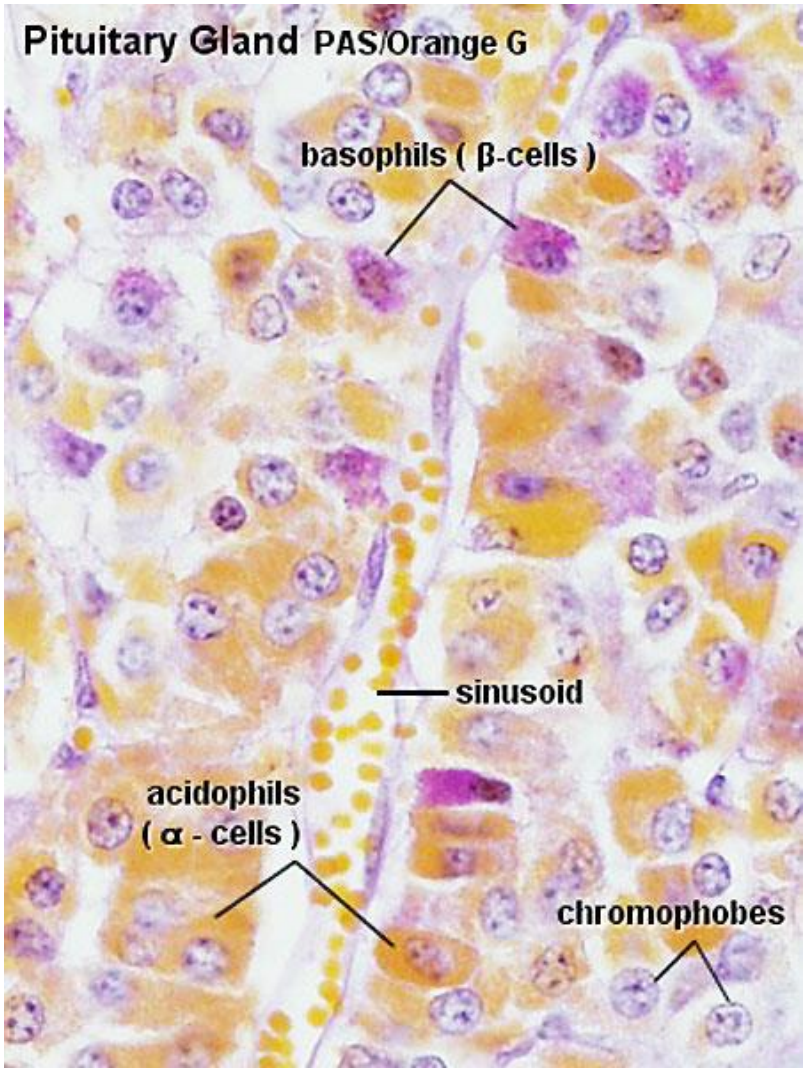


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Histology of pars distalis

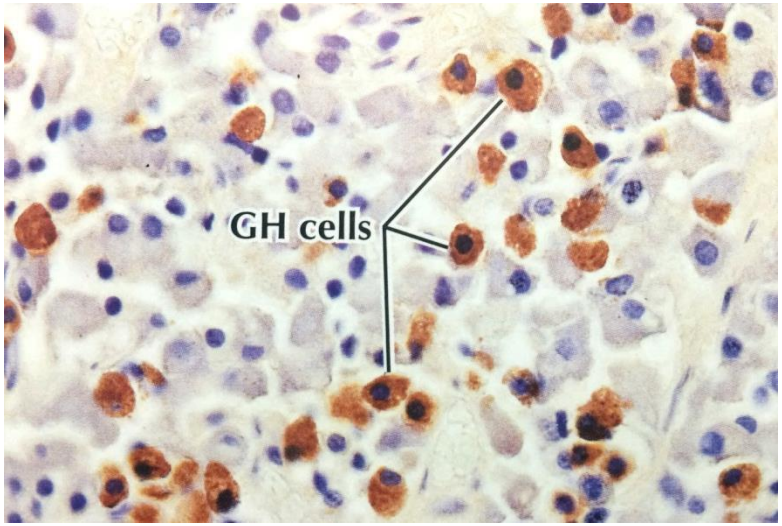
1. Chromophobes (~ 50%) : stain poorly with H&E and do not secrete hormones.
2. Chromophils : stain with H&E and secrete hormones which are Acidophils (~ 40%) and Basophils (~ 10%)





http://www.lab.anhb.uwa.edu.au/

Histology of pars distalis

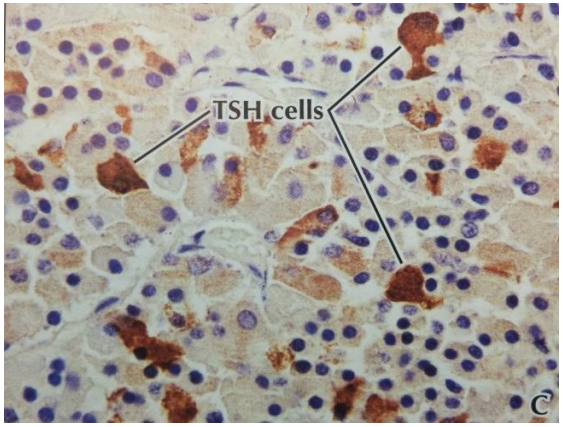


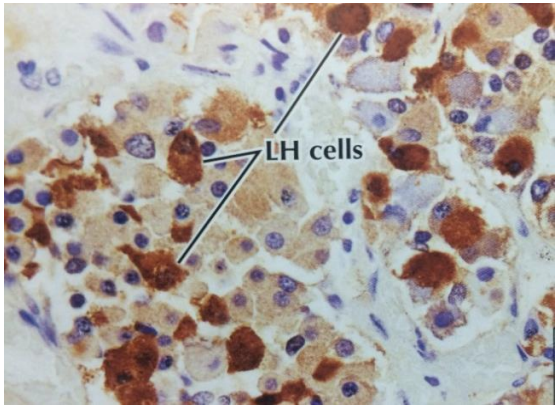
Netter's Essential Histology , page 221
Special staining with specific antibody

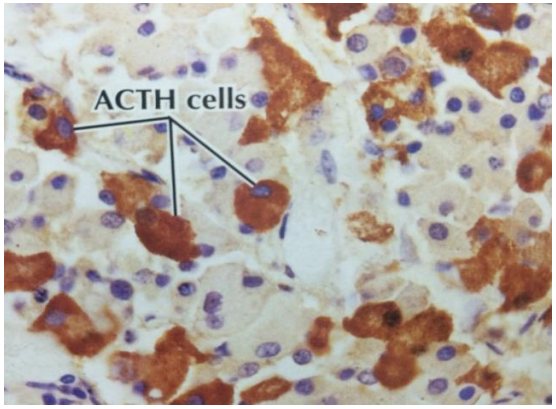
Acidophils (~ 40%)

1. **Somatotrophs** → growth Hr. (**GH**)
2. **Mamotrophs (Lactotrophs)** → prolactin (**PRL**)

Histology of pars distalis







Basophils (~ 10%)

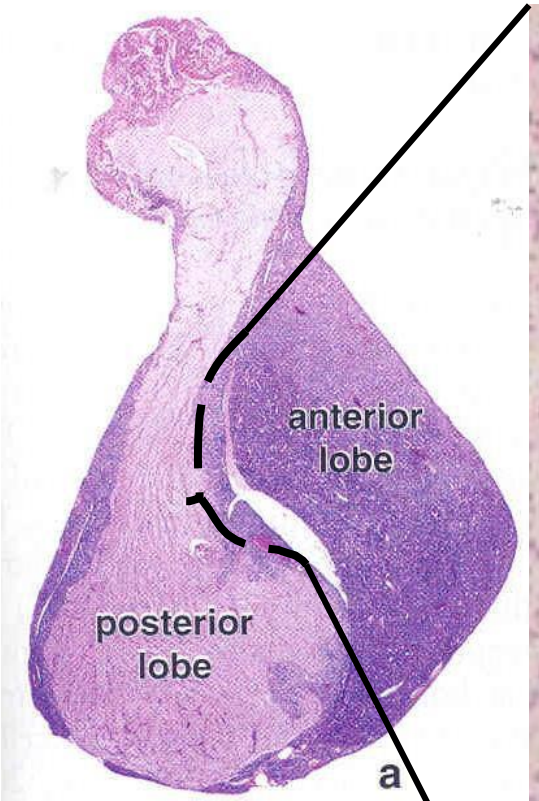
1. **Thyrotrophs** → thyroid-stimulating hr. (**TSH**)

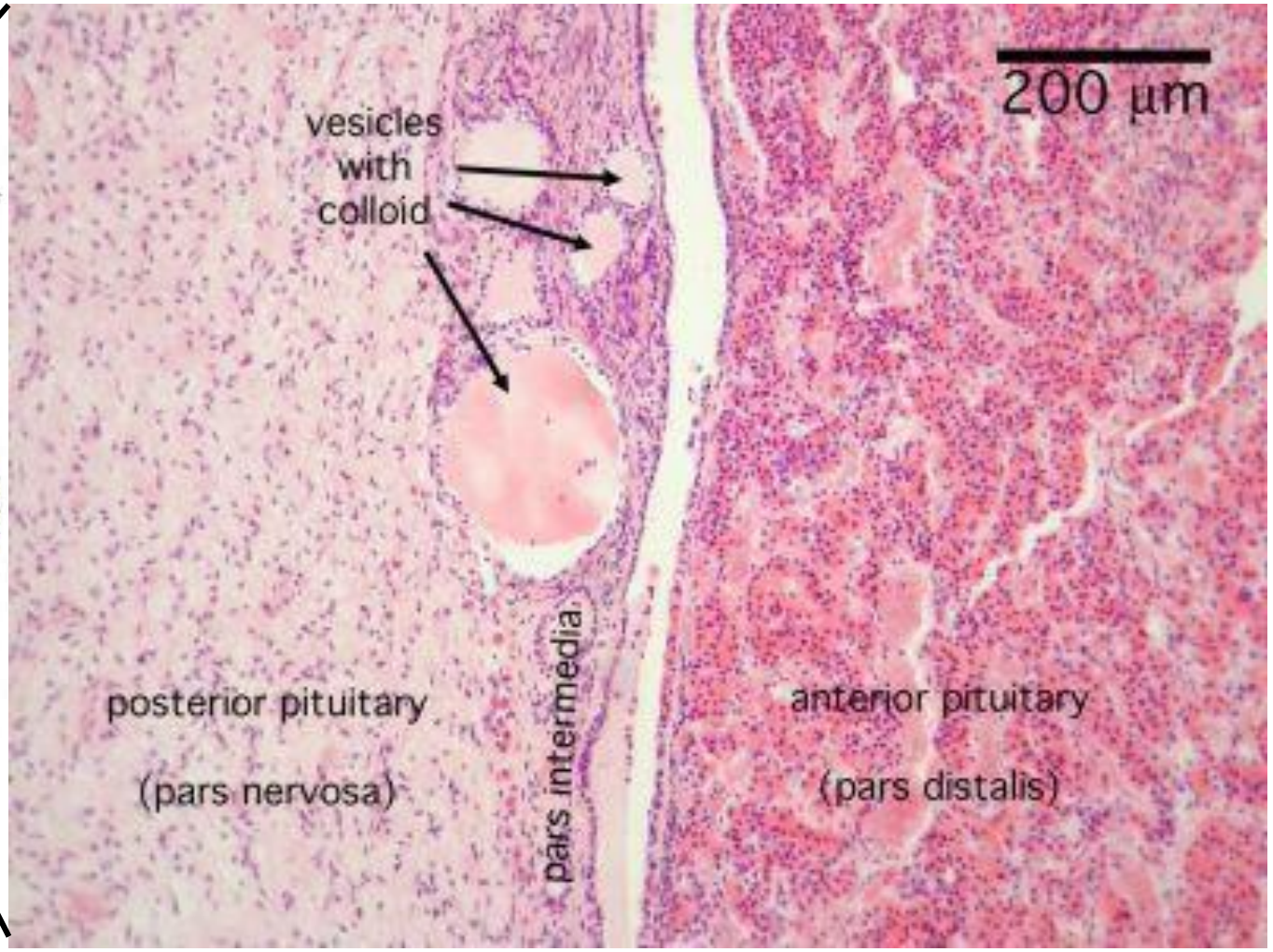
2. **Gonadotrophs** → follicle-stimulating Hr. (**FSH**) & luteinizing hr. (**LH**)

3. **Corticotrophs** →adrenocorticotropic hr. (**ACTH**)

Netter's Essential Histology, page 221
Special staining with specific antibodies

Histology of pars intermedia



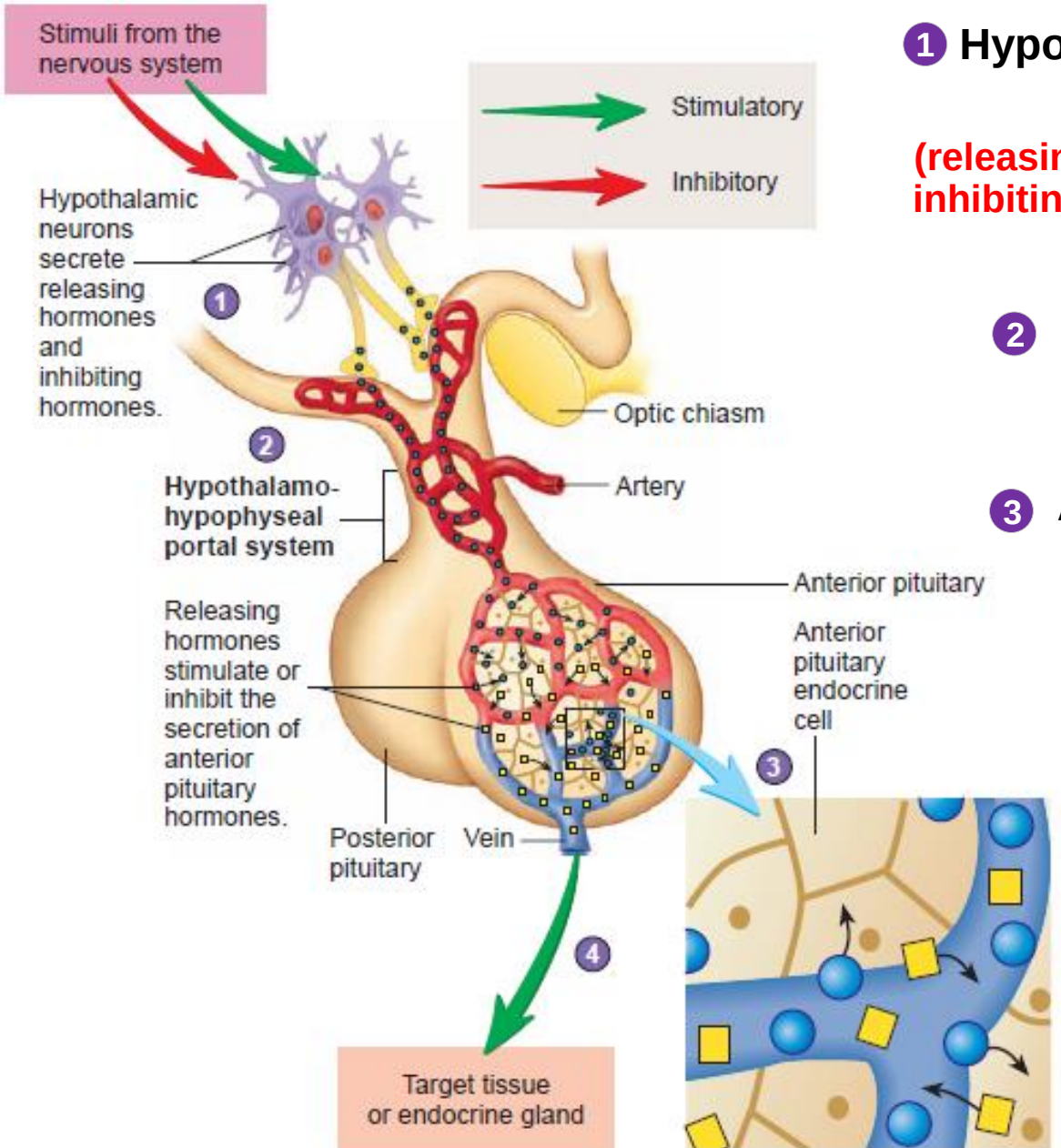


- **Corticotrophs (melanotrophs):** melanocyte-stimulating hormone (**MSH**) stimulates melanocytes to produce melanin pigment (amphibians & reptiles)
- Contains numerous colloid (protein)-filled cysts

*** Unknown function in human*

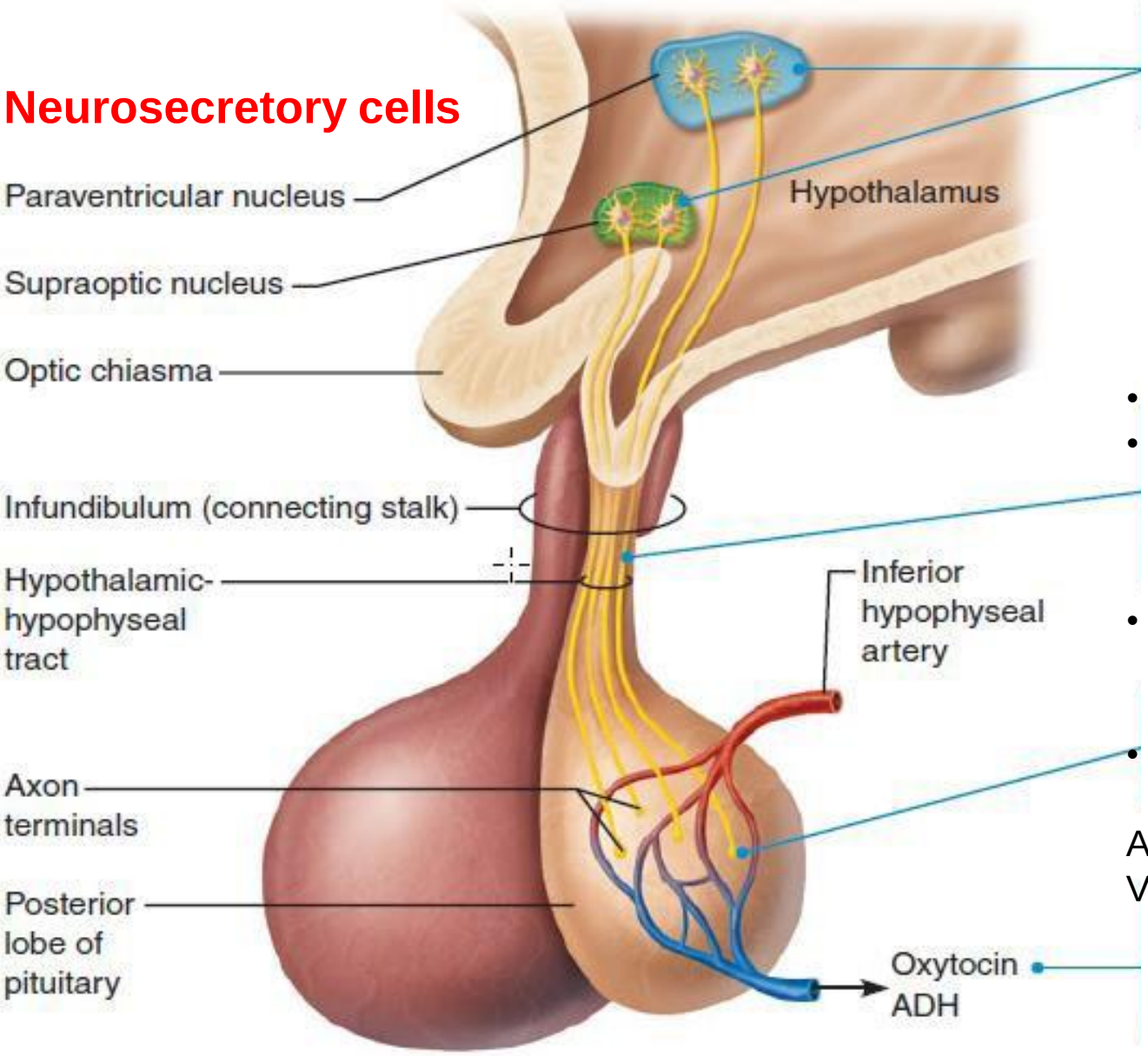
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Hypothalamic control of hormone secretion from the Anterior Lobe



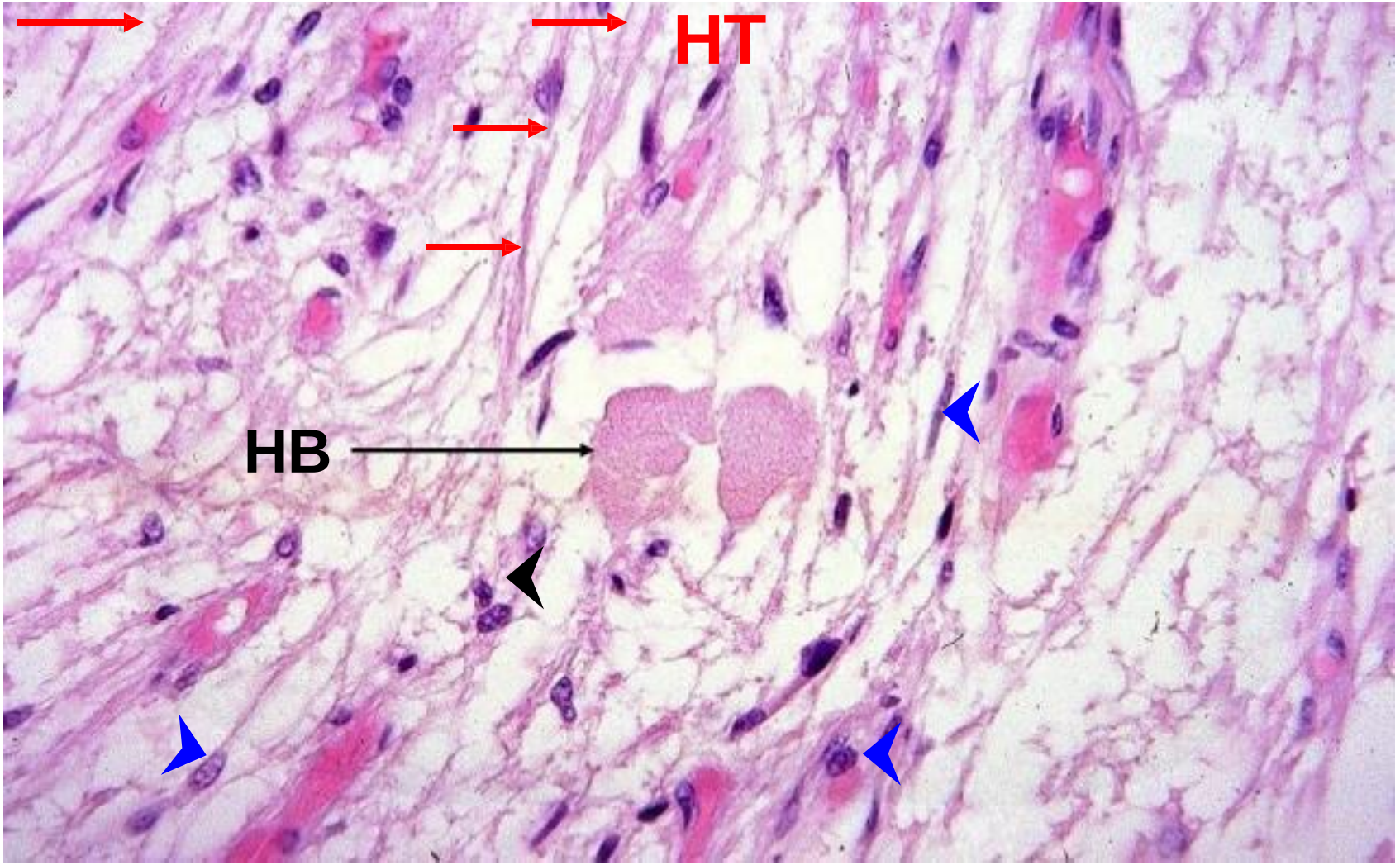
1. **Hypothalamic neurosecretory cells**
2. (releasing or inhibiting hr.) **Hypothalamic hormones**
3. Primary plexus of Hypophyseal portal system
4. **Ant. pituitary endocrine cells**
5. **Hormones**
6. Secondary plexus of Hypophyseal portal system
7. Vein
8. Target tissues

Functions of pars nervosa



- Structurally a part of the brain
- Its axons make up the hypothalamo-hypophyseal tract which arises from hypothalamus
- Axon terminals in the pars nervosa (Herring body) --> stored hormones
- Two nuclei
  1. **Supraoptic n.** → Antidiuretic h. (ADH) or Vasopressin
  2. **Paraventricular n.** → Oxytocin

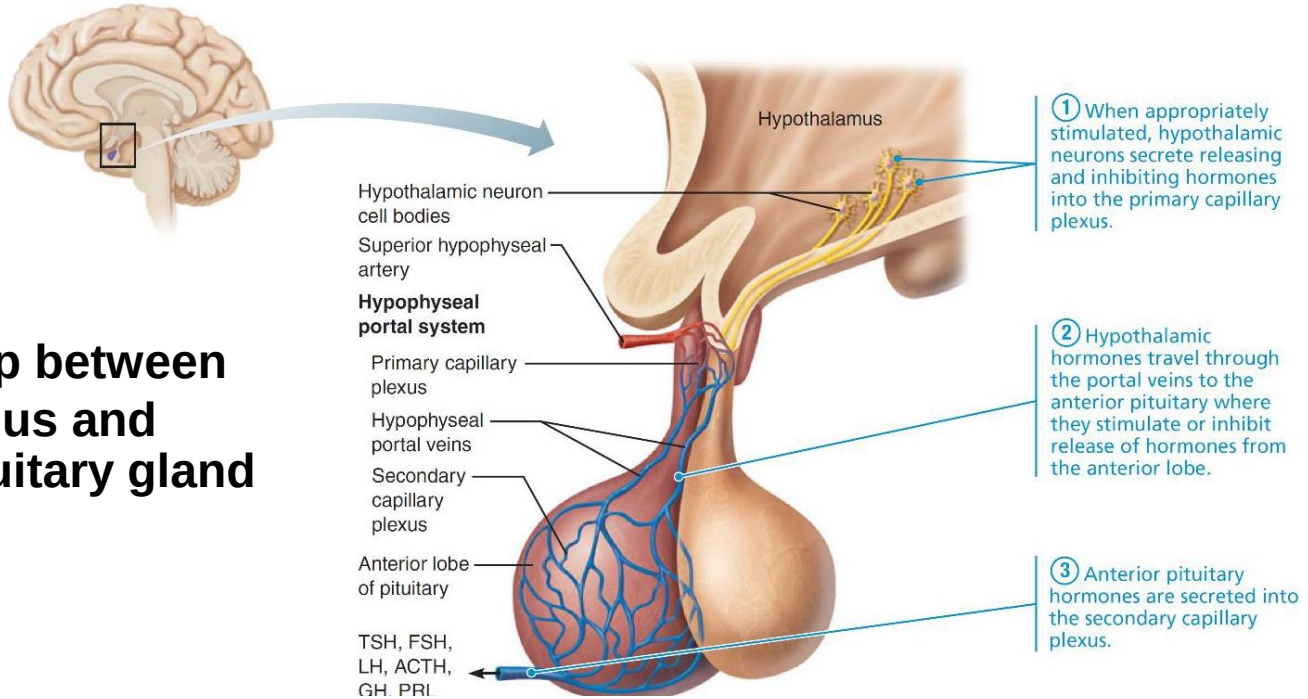
Histology of pars nervosa



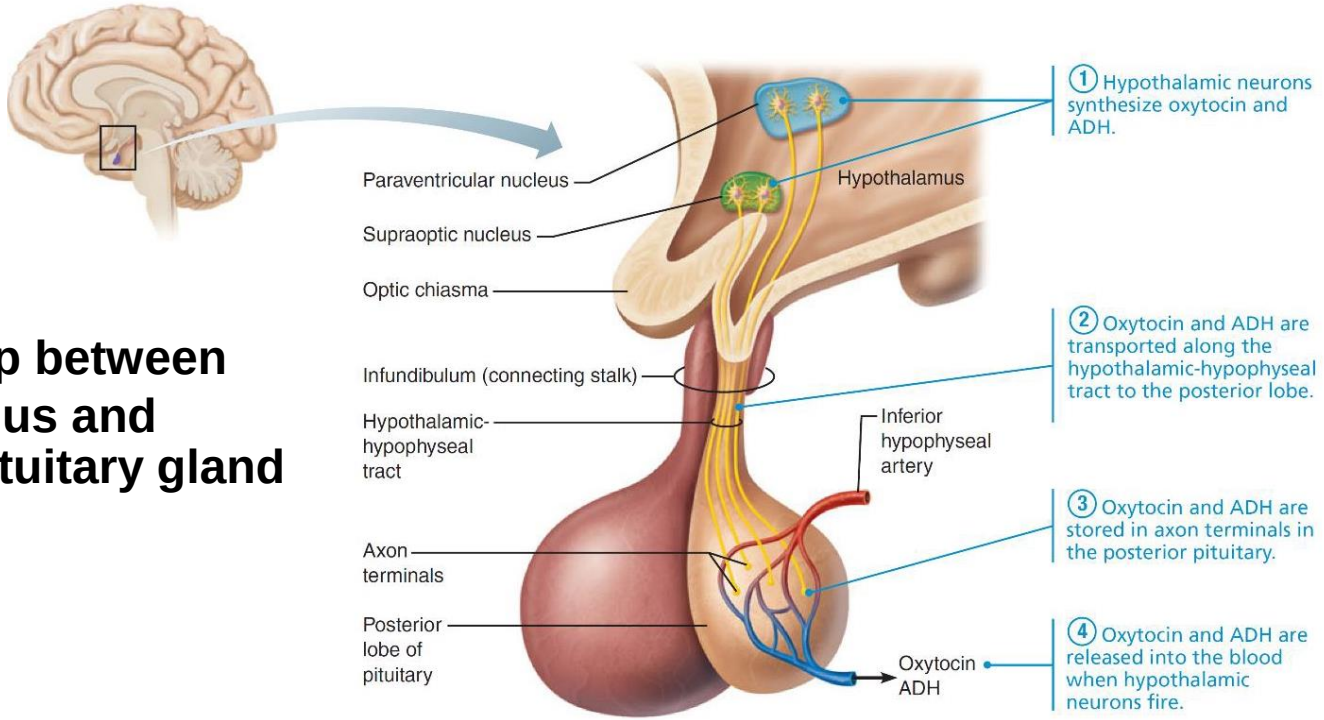
- Black arrows indicate **Herring bodies**: dilated terminal expansions of axons
- Arrowheads indicate **Pituicytes**: ensheath and support axons
- Red arrows indicates Axon bundles of the **hypothalamo-hypophyseal tract (HT)**

http://www.lab.anhb.uwa.edu.au/

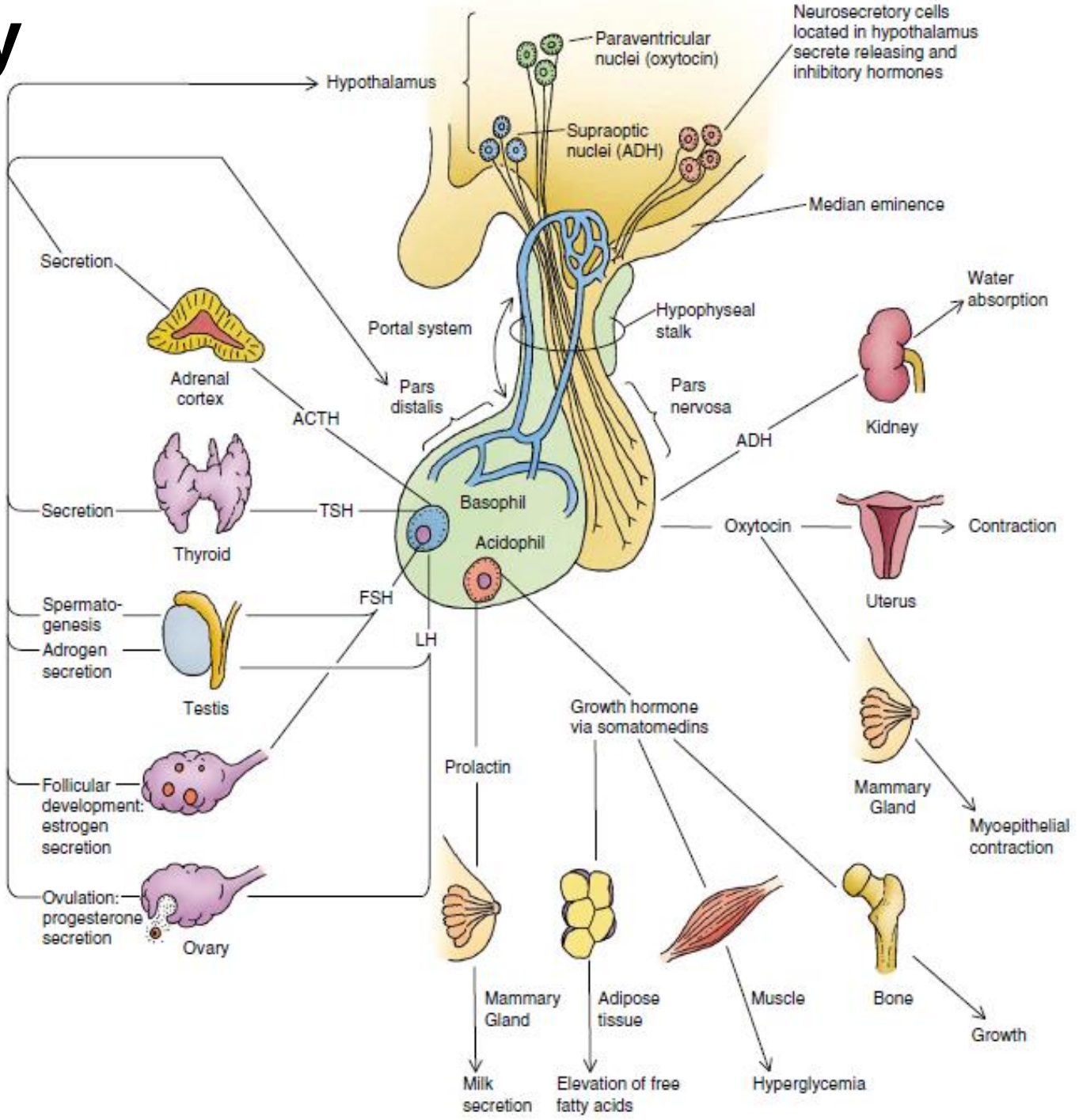
Relationship between hypothalamus and anterior pituitary gland



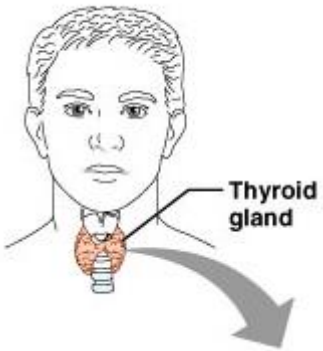
Relationship between hypothalamus and posterior pituitary gland

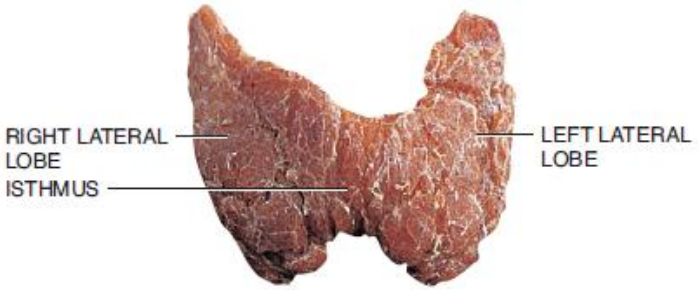


Summary

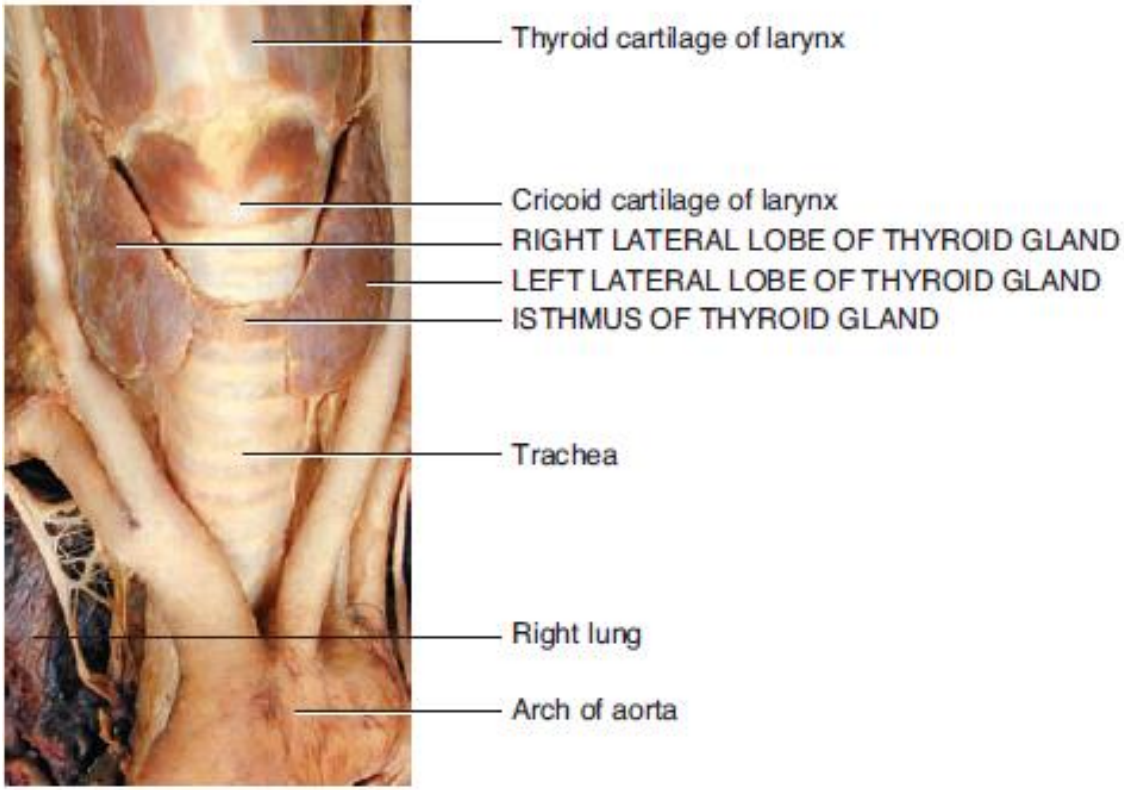

Hypothalamus
Paraventricular nuclei (oxytocin)
Neurosecretory cells located in hypothalamus secrete releasing and inhibitory hormones
Supraoptic nuclei (ADH)
Median eminence
Secretion
Adrenal cortex
Portal system
Hypophyseal stalk
Water absorption
Pars distalis
Pars nervosa
ACTH
ADH
Kidney
Secretion
TSH
Thyroid
Basophil
Acidophil
Oxytocin
Contraction
Uterus
Spermato-genesis
Adrogen secretion
FSH
LH
Testis
Growth hormone via somatomedins
Prolactin
Mammary Gland
Myoepithelial contraction
Follicular development: estrogen secretion
Ovulation: progesterone secretion
Ovary
Mammary Gland
Adipose tissue
Muscle
Bone
Growth
Milk secretion
Elevation of free fatty acids
Hyperglycemia

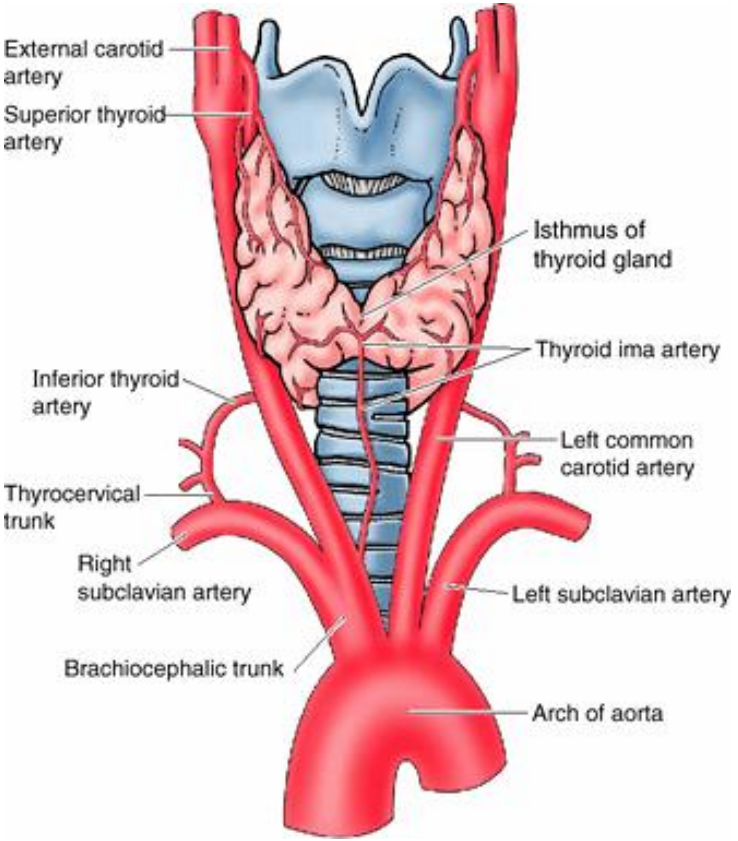
II. Thyroid gland





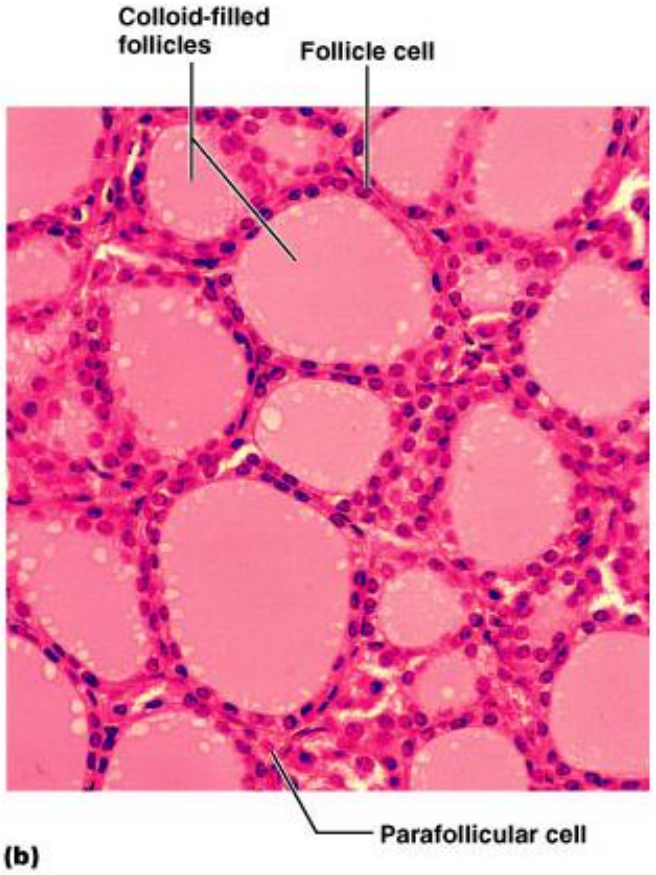
- Anterior neck on trachea just inferior to larynx
- Right & left lobes connected with isthmus
- Supplied with arterial blood from **superior thyroid artery**, a branch of the external carotid artery, and **inferior thyroid artery**, a branch of the thyrocervical trunk

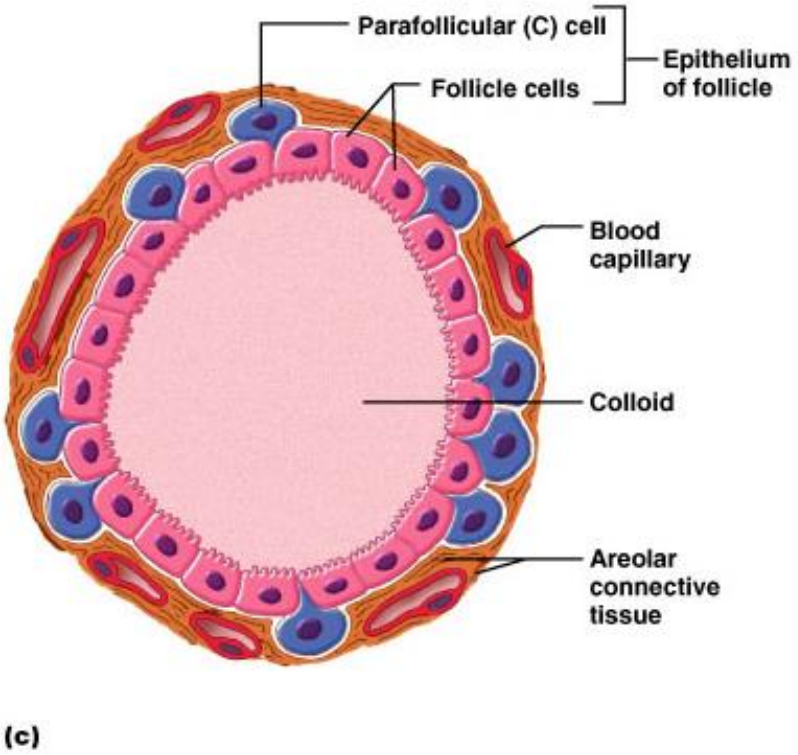




Histology of Thyroid Gland

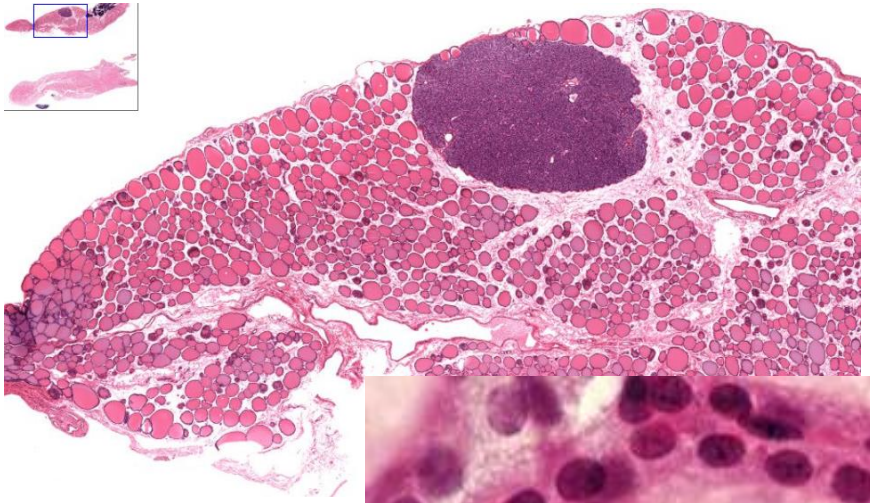
- **Follicular cells**: produce **thyroglobulin**, precursor of **thyroid hormones** (tyrosine based with 3 or 4 iodine molecule ➔ T4 (thyroxine) & T3
- Colloid lumen is of thyroglobulin
- **Parafollicular "C" cells**: produce **calcitonin** involved with calcium & phosphorus metabolism

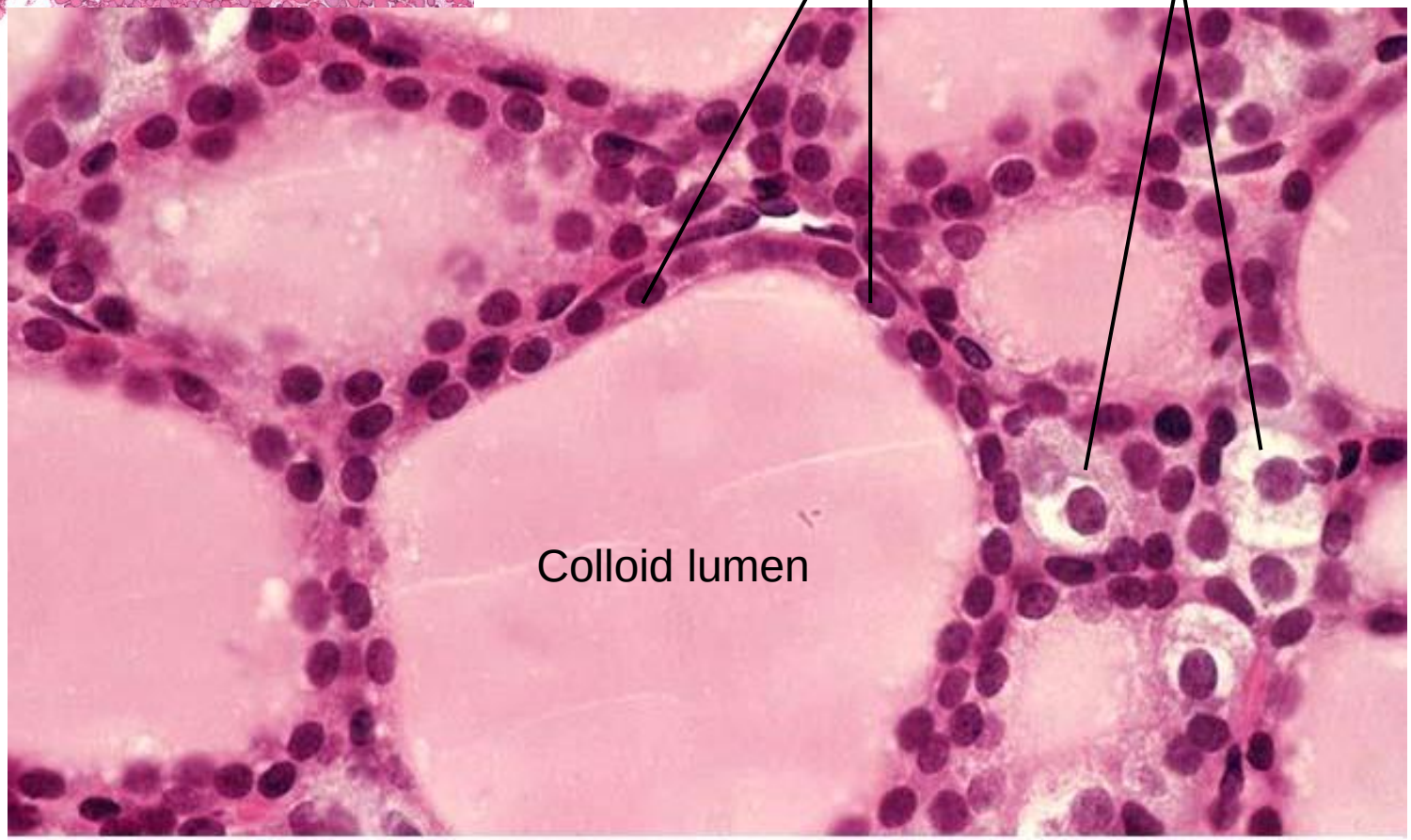




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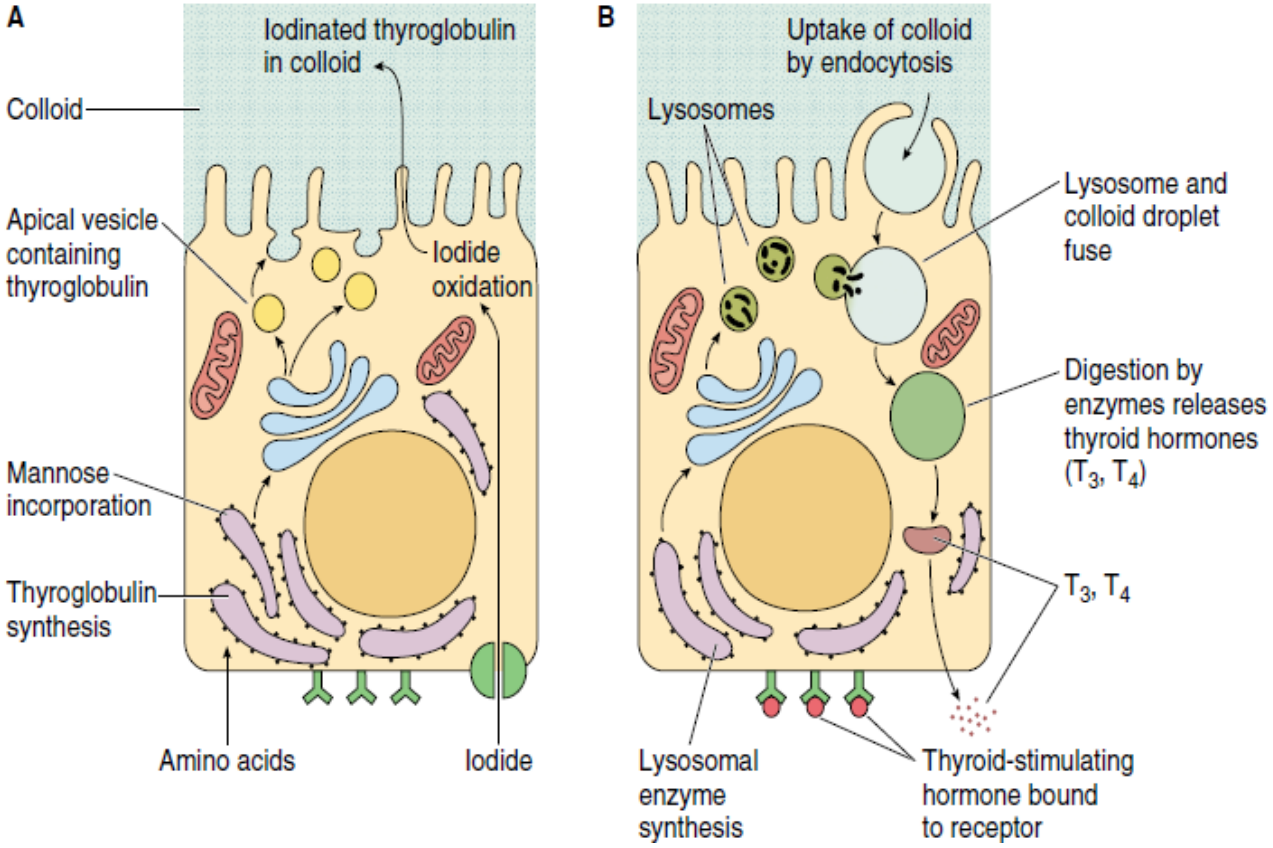
Histology of Thyroid Gland





http://www.histologyguide.com/slideview/MHS-209-thyroid-parathyroid-thymus/13-slide-1.html?x=0&y=0&z=-1&page=1

Thyroid hormone synthesis

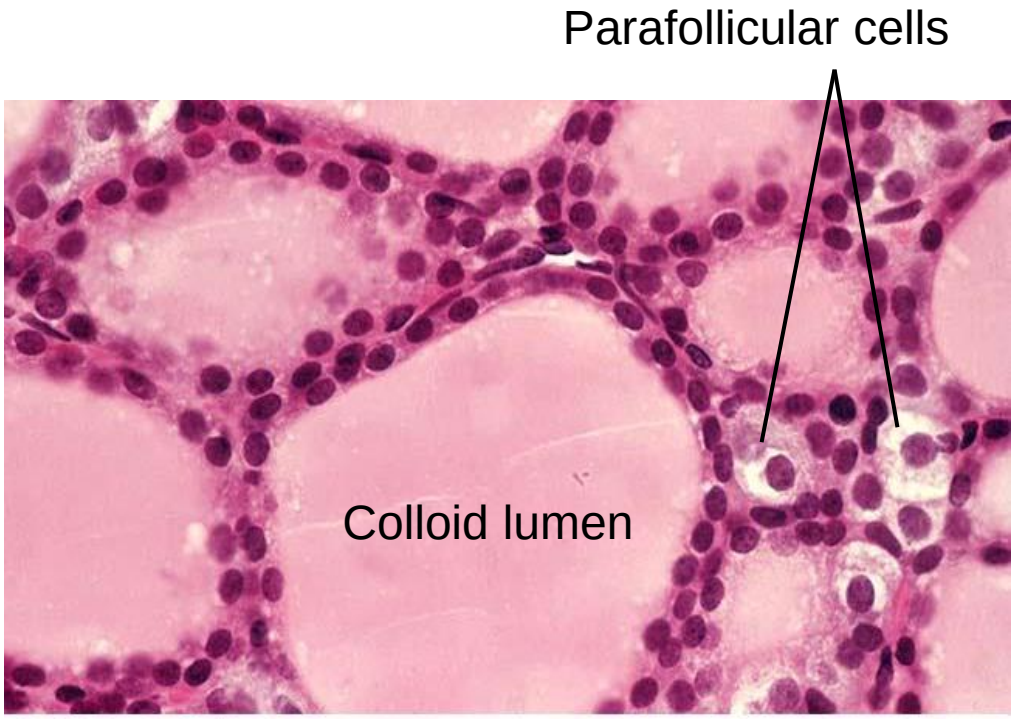


- Follicular cells produce thyroglobulin (TGB)
- Follicular cells also trap iodide & transport them from blood into cytosol

Thyroglobulin + iodine (I_2)
↓
colloid (T_1 &T_2)
↓
tri-iodothyronine (T_3)
tetra-iodothyronine (T_4)

- **Effects of thyroid hormones**
  → Increases the basal metabolic rate (**BMR**): the rate at which the body uses oxygen to transform nutrients (carbohydrates, fats and proteins) into energy
  → Affects many target cells throughout the body: protein synthesis, bone growth, neuronal maturation, cell differentiation

Parafollicular cells of thyroid gland

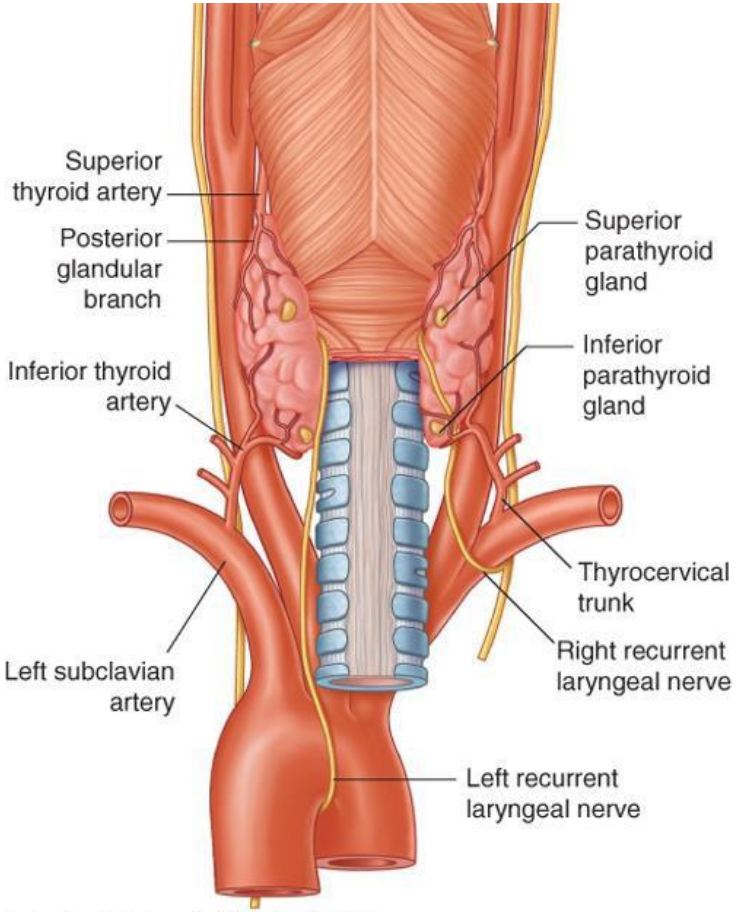


http://www.histologyguide.com/slideview/MHS-209-thyroid-parathyroid-thymus/13-slide-1.html?x=0&y=0&z=-1&page=1

- **Parafollicular** cells or **C** cells
- A few cells lie between follicles
- Larger than follicular cells & centrally located nuclei with clear cytoplasm
- Secrete **calcitonin** hr. when blood Ca^{2+} are high → **lowers Ca^{2+}** by **inhibiting** the action of osteoclasts in bone & increasing Ca^{2+} excretion by the kidney

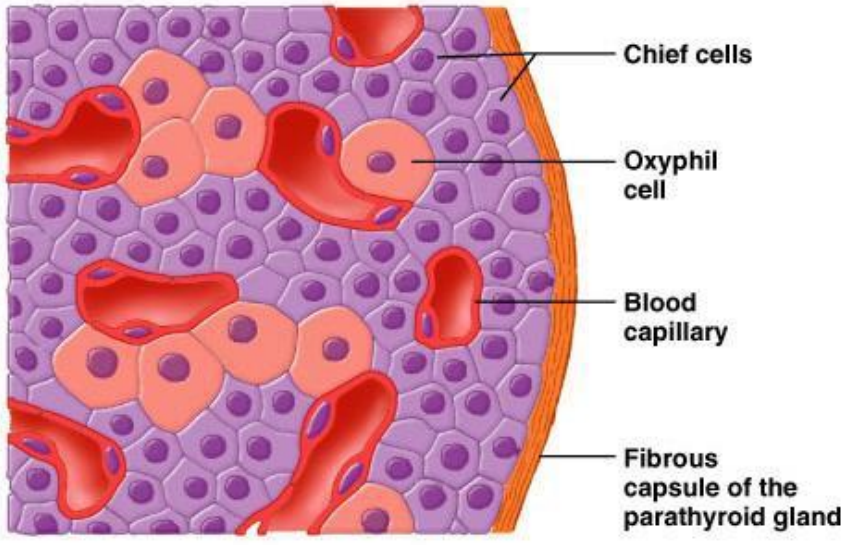
III. Parathyroid gland



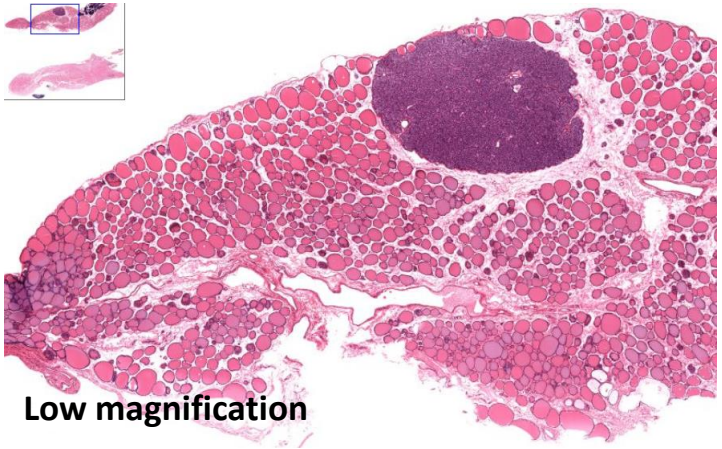


Drake: Gray's Anatomy for Students, 2nd Edition.
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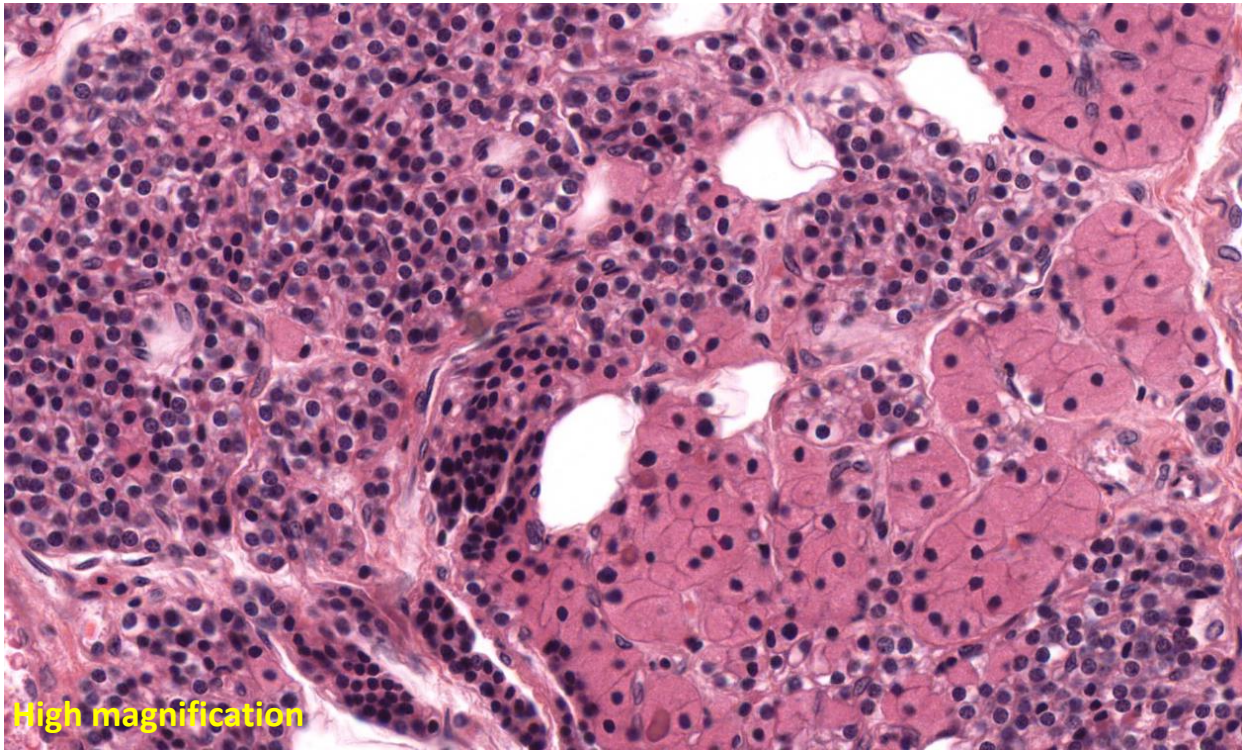
- Small, round masses of tissue, about 40 mg.
- Posterior surface of thyroid gland
- Contain **2 kinds** of cells:
  - **Chief cells → parathyroid hormone (PTH)** → the major regulator of the level of calcium, magnesium, & phosphate ions in the blood
  - Oxyphil cells: unknown function



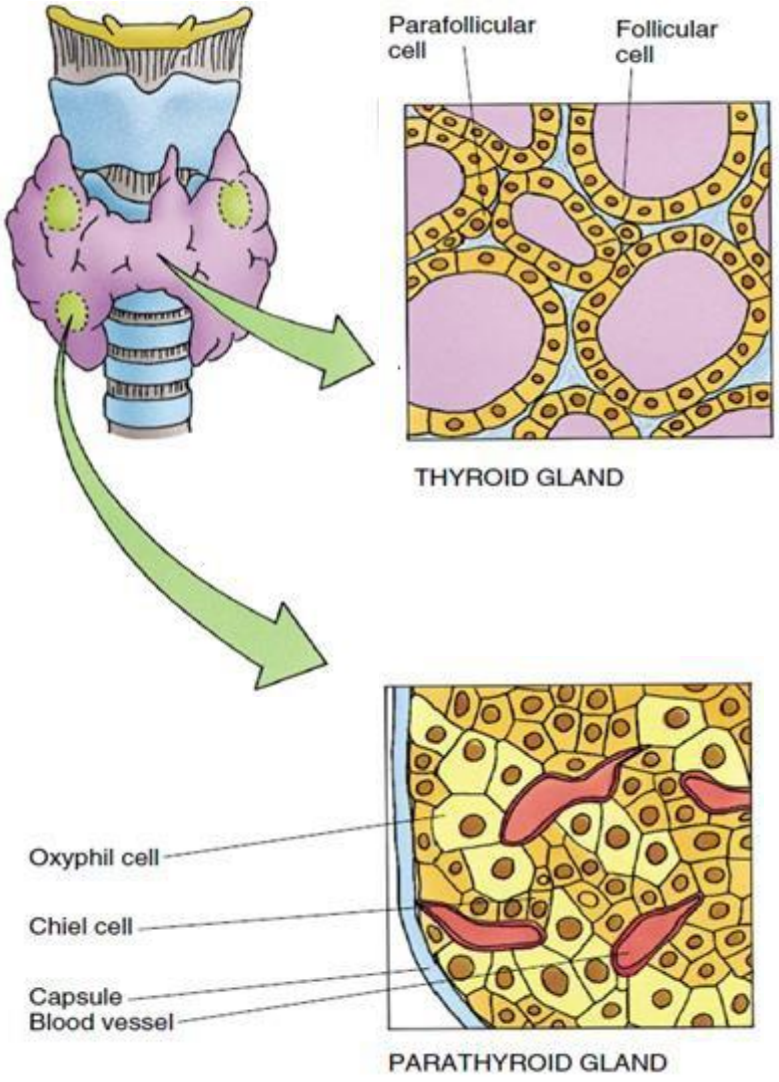
Histology of parathyroid gland

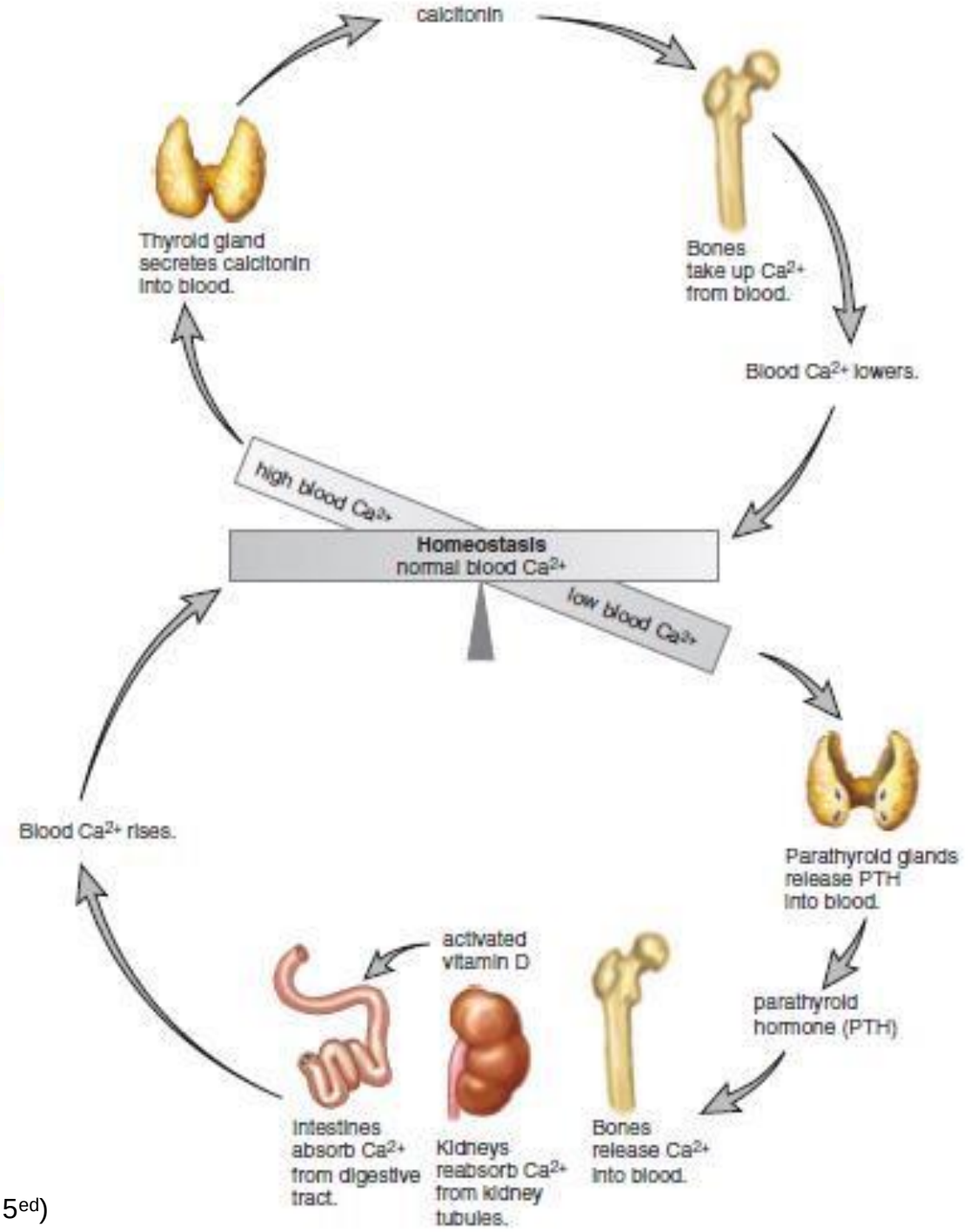


- **Increases blood Ca^{2+} (calcium) concentration when it gets too low**
- **Primary mechanisms of raising blood calcium**
  1. Stimulates osteoclasts to release more $Ca^{2+}$from bon
  2. Decreases secretion of Ca^{2+} by kidney
  3. Activates Vitamin D, which stimulates the uptake of Ca^{2+} from the intestine



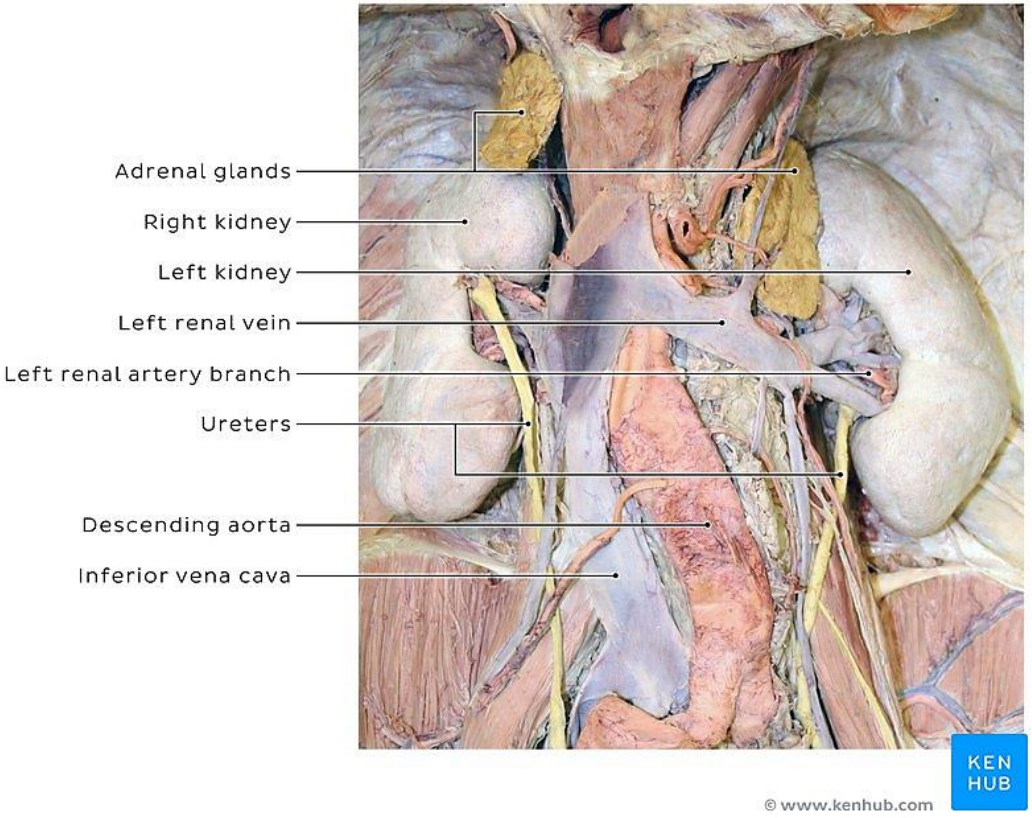
- Has opposite effect with calcitonin (which lowers Ca^{2+} levels)
- Unwitting removal during thyroidectomy was lethal





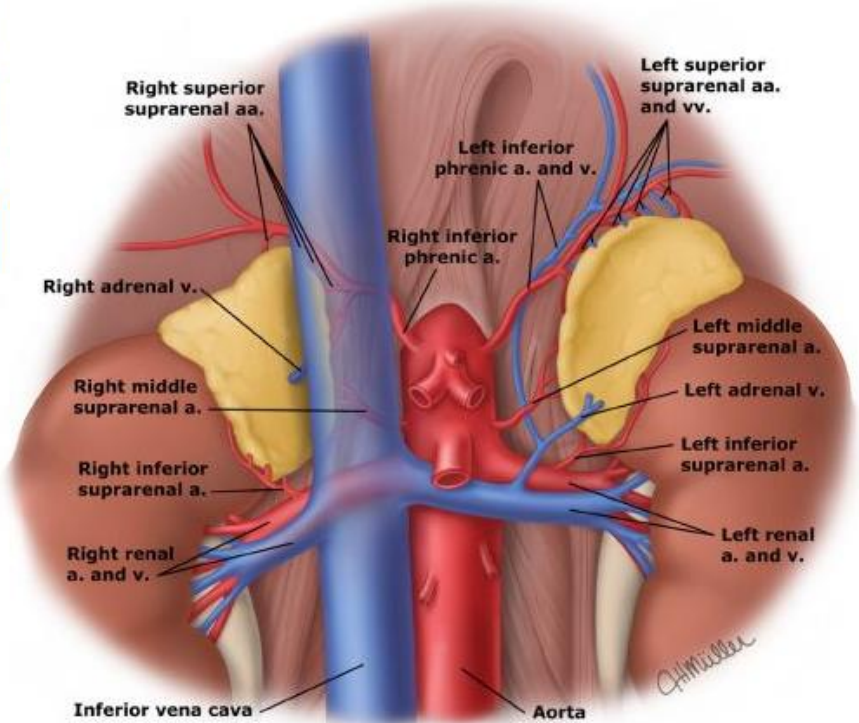
(Understanding human anatomy and physiology, 2004, 5ed)

IV. adrenal (suprarenal) gland



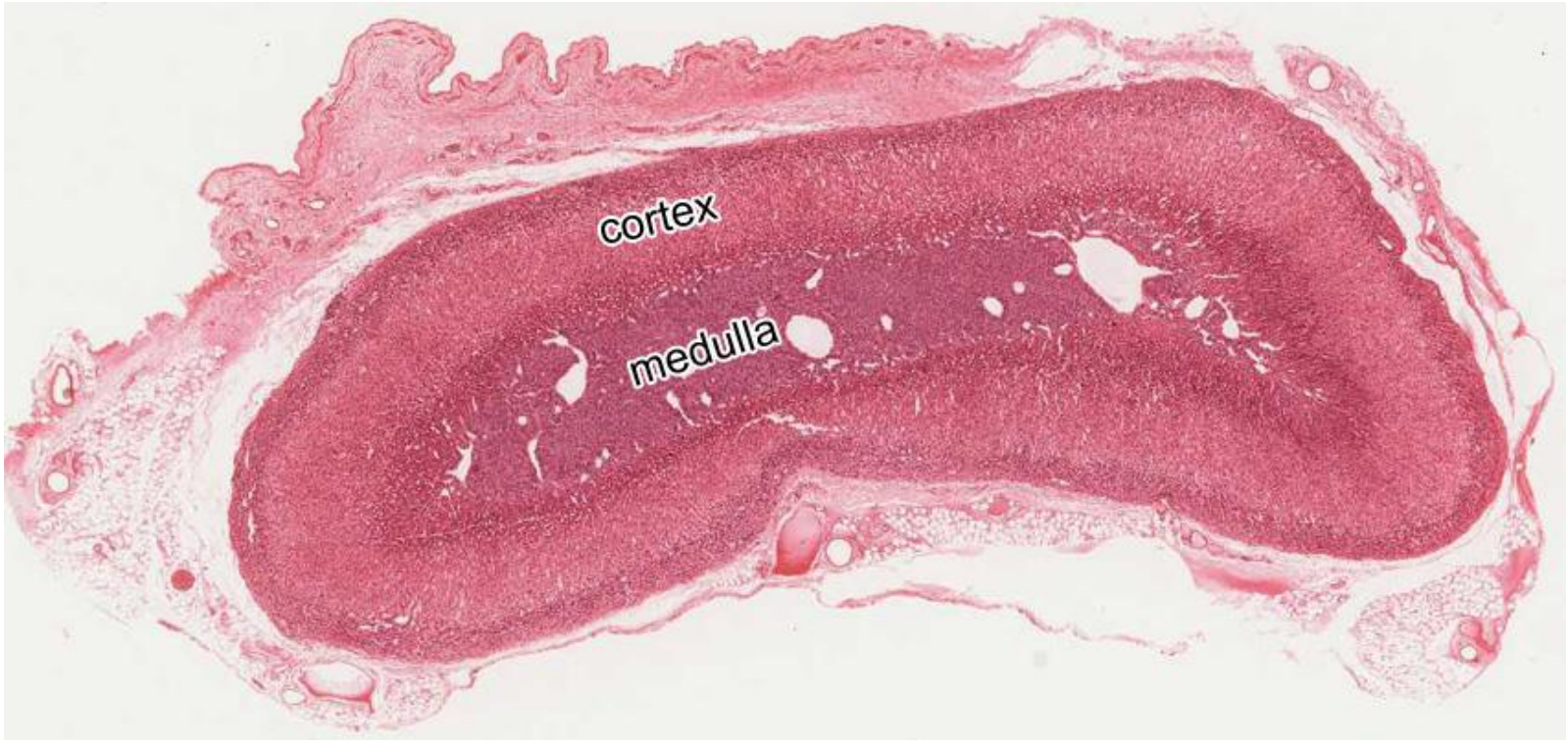
A flattened pyramidal shape that lies superior to each kidney in retroperitoneal space

Three main groups of vessels supply the adrenal gland: (i) **superior suprarenal arteries**, derived from the inferior phrenic arteries; (ii) **middle suprarenal artery**, derived from the aorta; and (iii) **inferior suprarenal renal arteries**, derived from the adjacent renal artery.

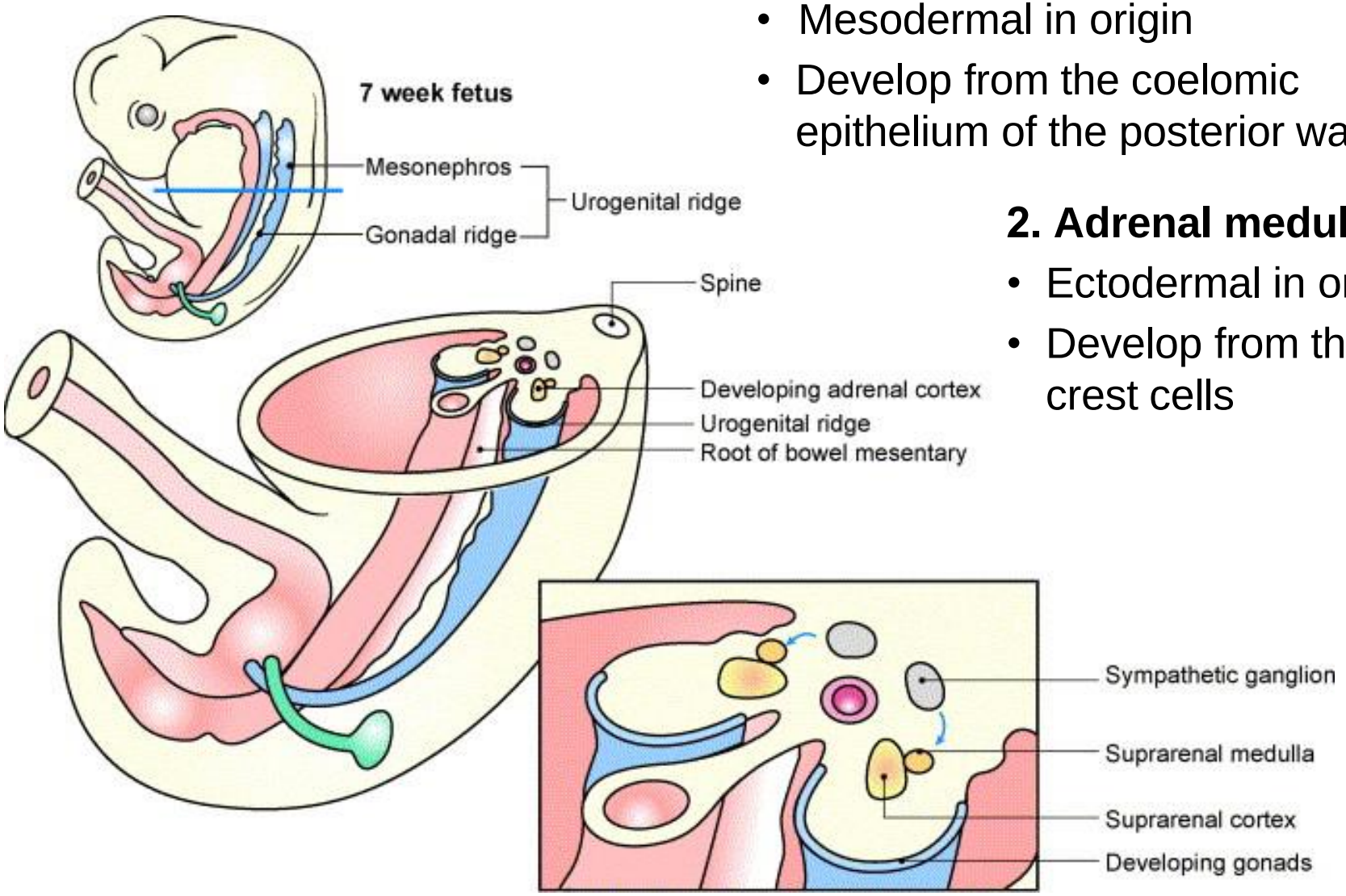


Internal structure of the adrenal gland

- Adrenal capsule – collagen & connective tissue
- **Adrenal cortex** -- peripherally located
- **Adrenal medulla** -- centrally located



Two parts of adrenal gland develop from two different origins



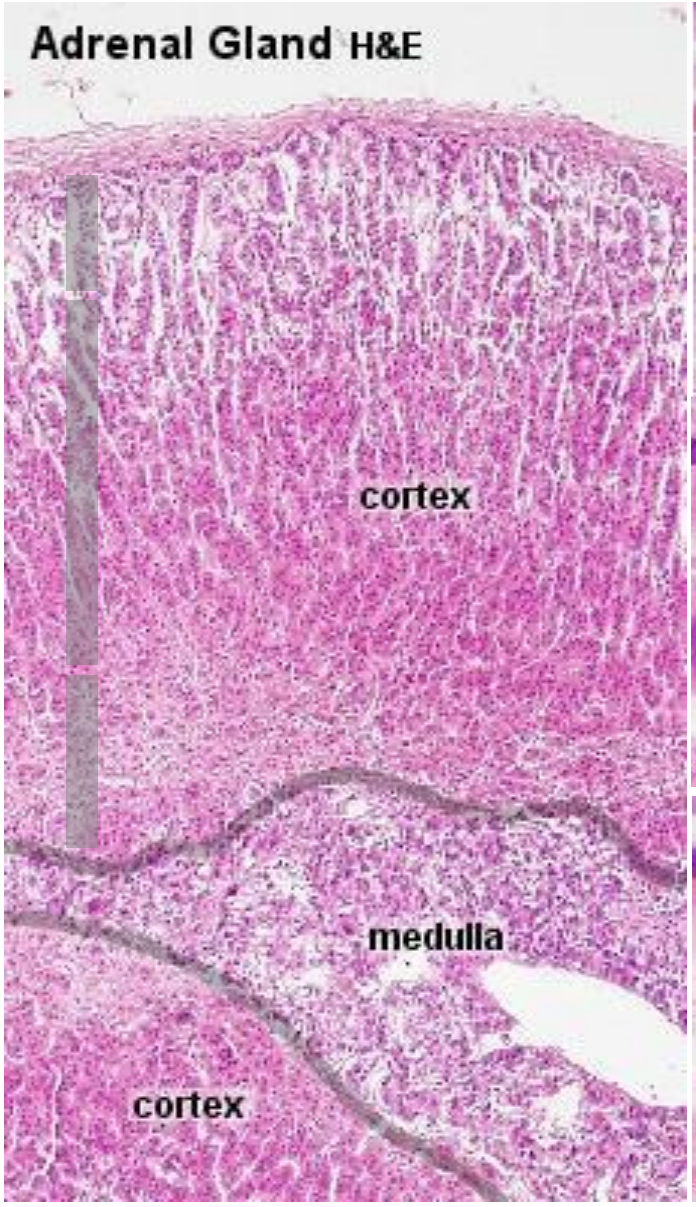
1. Adrenal cortex:

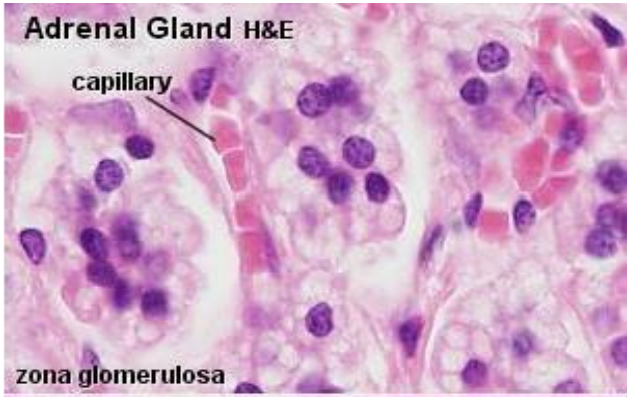
- Mesodermal in origin
- Develop from the coelomic epithelium of the posterior wall

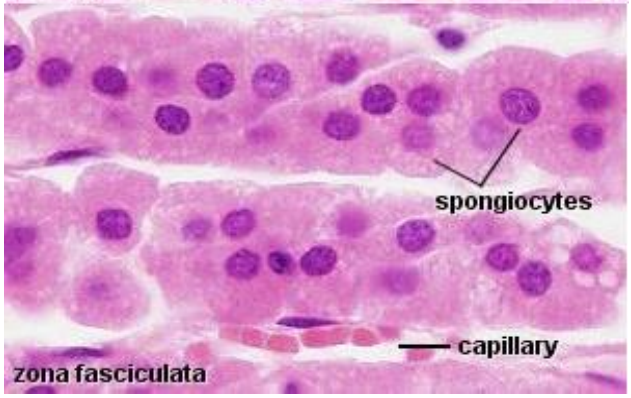
2. Adrenal medulla:

- Ectodermal in origin
- Develop from the neural crest cells

Histology of adrenal cortex









Zona glomerulosa:

- glomerular-like clusters of cells
- secretes mineralocorticoids, mainly **aldosterone**

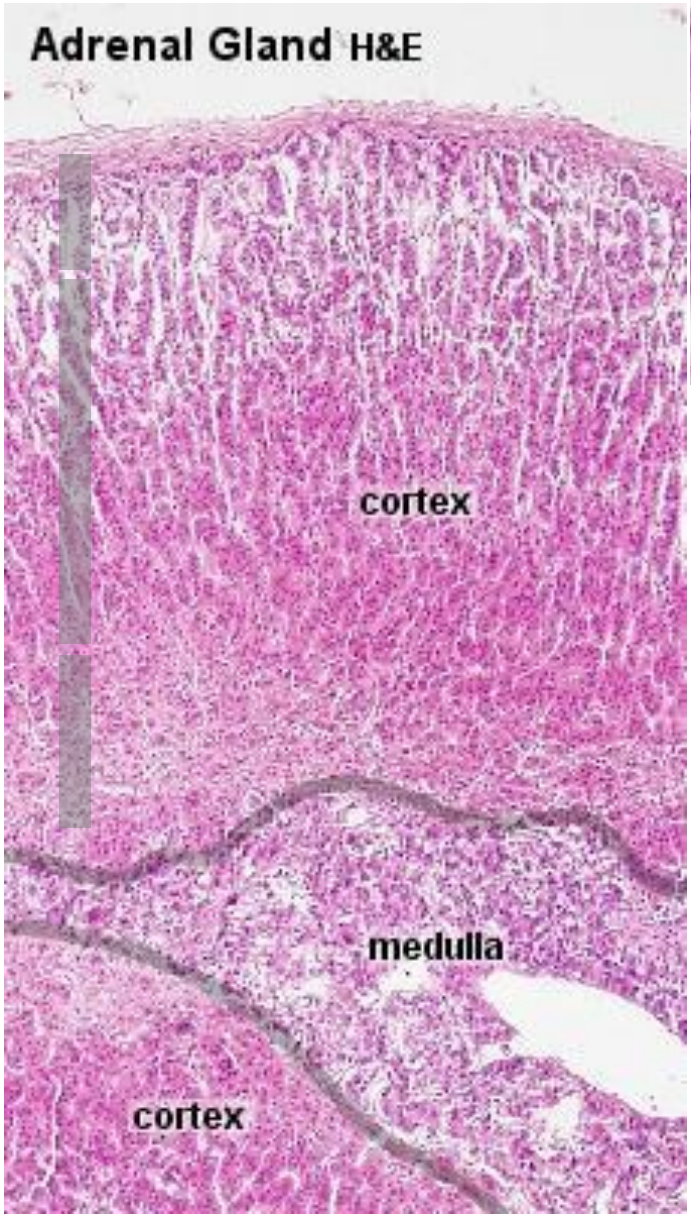
Zona fasciculata:

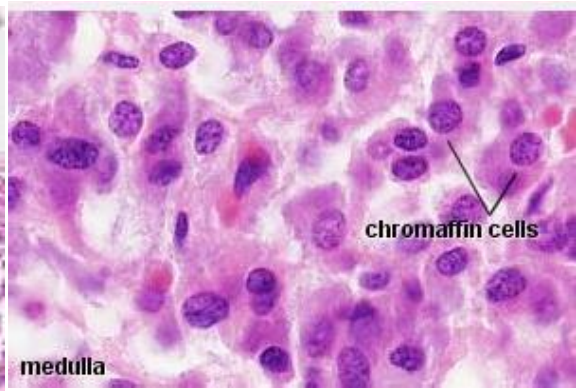
- two-cell wide vertical cords
- secretes glucocorticoids, mainly **cortisol**

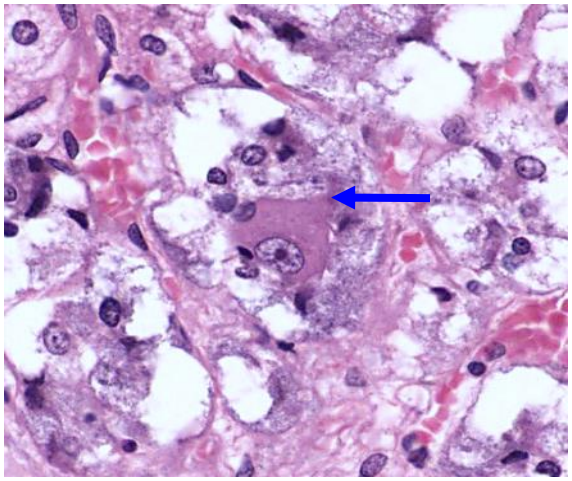
Zona reticularis:

- Branching network
- secretes small amounts of weak androgens

Histology of adrenal medulla

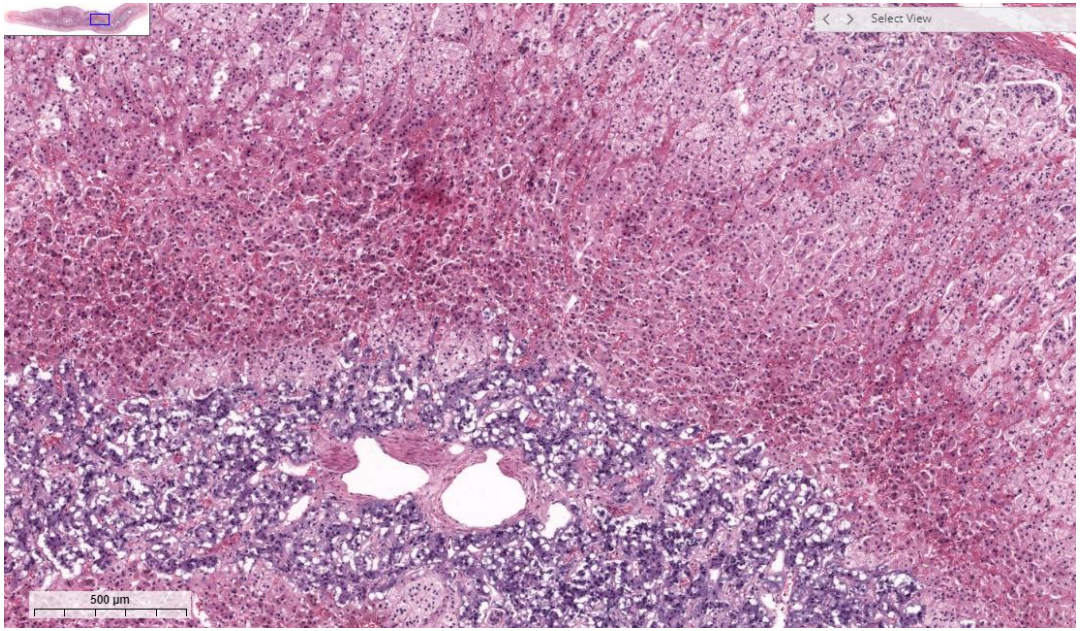


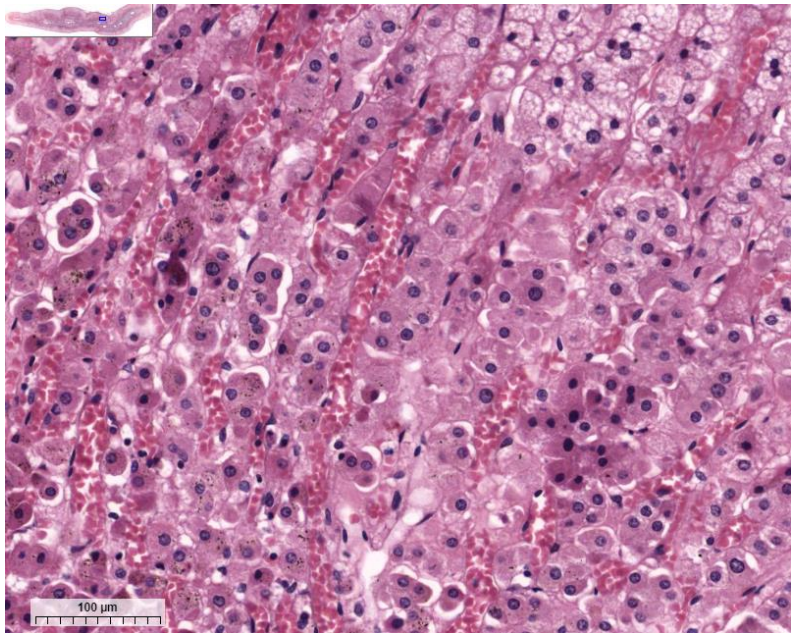




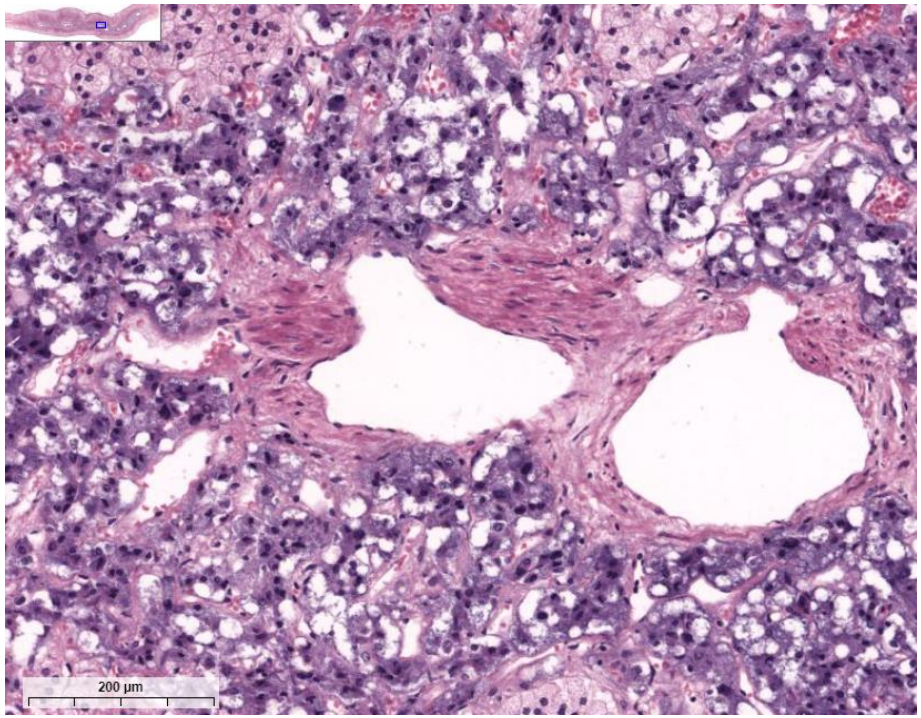
Adrenal medulla (modified post-sympathetic ganglia of the ANS):

- **Chromaffin cells** secrete epinephrine and norepinephrine (NE)
- Ganglion Cells - infrequent sympathetic ganglion cells.





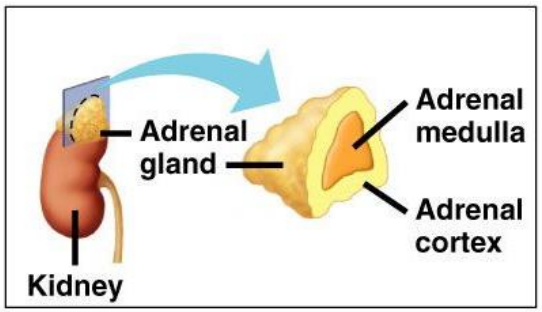
Sinusoidal capillaries - rich network of blood vessels.



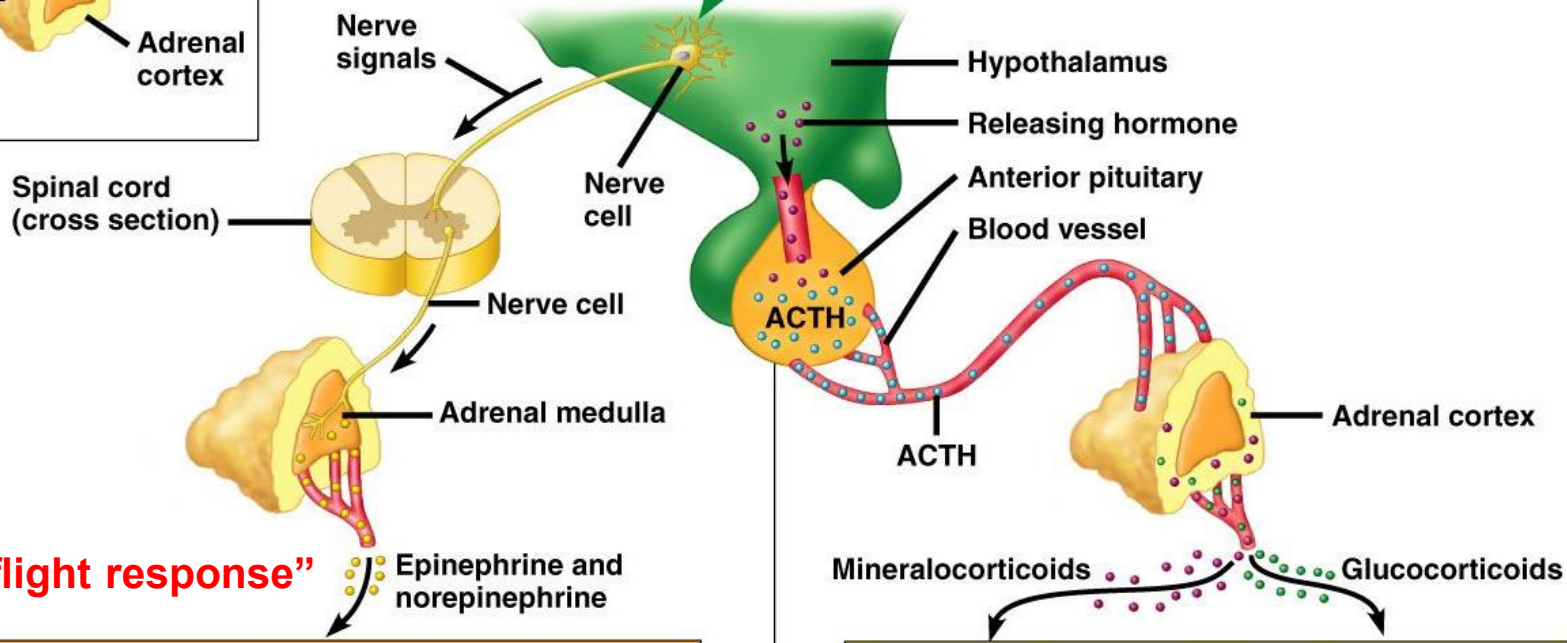
Medullary vessels - large veins that drain the organ.

http://www.histologyguide.com/slidebox/13-endocrine-glands.html?fbclid=IwAR1u6r11MgfgdQp8Mp7b589ctuLQo8ABwY9KGLkRDIBerj9_i9pz9oIa-tk

Stress and Adrenal glands







"The fight-or-flight response"

Short-term stress response

1. Glycogen broken down to glucose; increased blood glucose
2. Increased blood pressure
3. Increased breathing rate
4. Increased metabolic rate
5. Change in blood-flow patterns, leading to increased alertness and decreased digestive and kidney activity

Long-term stress response

Mineralocorticoids

1. Retention of sodium ions and water by kidneys
2. Increased blood volume and blood pressure

Glucocorticoids

1. Proteins and fats broken down and converted to glucose, leading to increased blood glucose
2. Immune system may be suppressed

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Summary

1. Mineralocorticoids: mainly **Aldosterone** → homeostasis of Na^+ and K^+ → increase reabsorption by distal convoluted tubule of kidneys

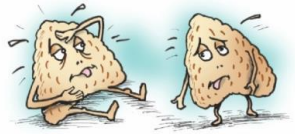
2. Glucocorticoids: mainly **Cortisol**

- Helps the body deal with **stressful** situations within minutes
  - Physical: trauma, surgery, exercise
  - Psychological: anxiety & depression
  - Physiological: fasting, hypoglycemia, fever, infection
- Regulate metabolism : protein breakdown, glucose formation, & lipolysis
- Anti-inflammatory effects inhibit white blood cells in inflammatory response

3. Androgens: mainly dehydroepiandrosterone or DHEA → assist in early growth of axillary & pubic hair in both genders

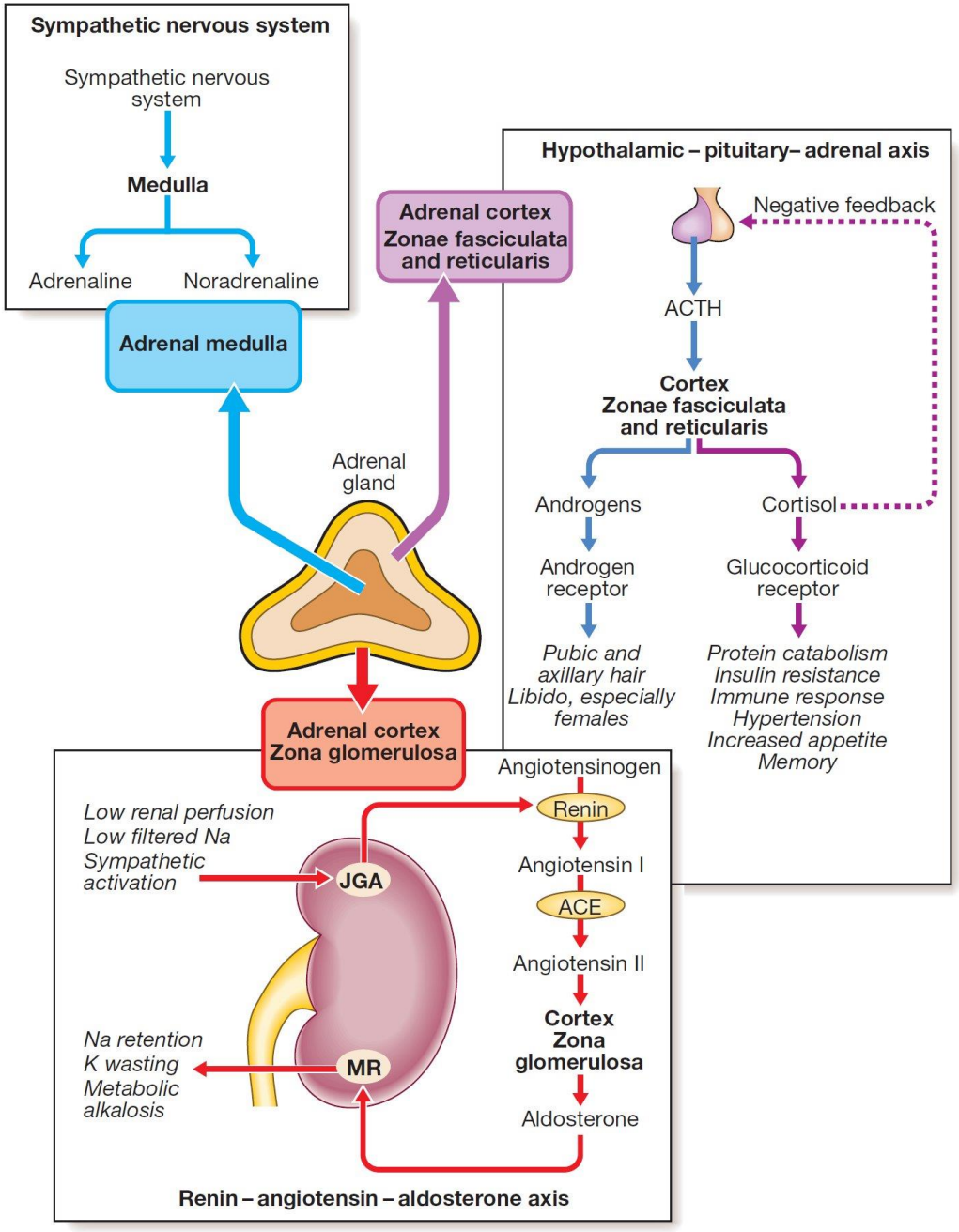
4. Epinephrine & Norepinephrine

- These hormones increases heart rate, blood pressure, blood glucose levels, & blood flow into the heart and lungs so the body can respond to the initial stress.



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Structure and function of the adrenal glands. (ACE = angiotensin-converting enzyme; JGA = juxtaglomerular apparatus; MR = mineralocorticoid receptor)

Source : Davidsons Essentials of Medicine, 2e